Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

Anonymous submission

Abstract

Political information spreads through social media via both sharing and conversation, and platforms differ in how these processes unfold. This study examines candidate-centered political diffusion about the 2024 U.S. presidential election on Truth Social and Bluesky, two ideologically distinct but technically comparable platforms. Using repost and reply networks constructed from public posts mentioning Biden or Trump during May–June 2024, we analyze cascade structure and evaluate the roles of influencers, topic variation, ideological composition, and interaction alignment. Repost diffusion is broadly similar across platforms, but reply cascades differ substantially: interactions on Truth Social are wider and shallower, whereas Bluesky supports narrower but deeper conversational threads. These differences are statistically accounted for more strongly by ideological composition and alignment than by influencers or topic variation. Fine-grained motif analysis further reveals asymmetric interaction patterns, including distinct roles for left-, right-, and center-leaning users across platforms. Overall, the findings suggest that variation in election-related information diffusion reflects not only who participates, but also how interaction patterns organize opportunities for cross-group engagement, with replies being especially sensitive to ideological structure.

Introduction

Social media platforms facilitate information cascades, chains of reposts, replies, and other interactions through which political content circulates online (Easley and Kleinberg 2010; Gruhl et al. 2004; Guille et al. 2013). Understanding how these cascades form has become a central concern across computational social science, communication, and network analysis. Past research has shown how different types of content, such as videos, petitions, or news, spread on platforms including Twitter, Facebook, and LinkedIn, and how false news in particular can diffuse differently than true news (Goel et al. 2016; Cheng et al. 2014; Anderson et al. 2015; Vosoughi, Roy, and Aral 2018; Zhao et al. 2020).

Much of this work highlights two broad influences. First are platform affordances, features like reply visibility, feed ranking, or sharing mechanisms, that determine what users see and how they can interact (Boyd 2010; Treem and Leonardi 2013). Second are community dynamics: who participates, how like-minded or diverse those users are, and whether they engage across ideological boundaries (Conover et al. 2011; Garimella et al. 2018). These factors together shape whether cascades remain shallow amplifications or develop into extended conversations.

Existing studies have often aggregated cascades by content type (e.g., true vs. false news) and examined size, depth, breadth, or virality to identify broad drivers such as novelty, veracity, or emotional tone (Goel et al. 2016; Vosoughi, Roy, and Aral 2018; Berger and Milkman 2012; Zollo et al. 2015; González-Bailón et al. 2024; Slaughter et al. 2025). However, most analyses focus on a single platform, limiting what we know about which aspects of diffusion are universal and which are platform-specific, especially in newer, ideologically distinct online environments (Cinelli et al. 2021).

This gap is increasingly relevant because the social media ecosystem has fractured. After the January 6 Capitol attack, Donald Trump launched Truth Social as a right-leaning alternative to mainstream platforms. More recently, Twitter's rebranding as X and its loosening of moderation have spurred many centrist and left-leaning users to migrate to Bluesky (Gerard, Botzer, and Weninger 2023; Failla and Rossetti 2024; Quelle and Bovet 2025). Although both platforms mimic Twitter's design, follower networks, reposts, and reply mechanics, their contrasting community compositions create distinct social contexts. Prior studies of Gab, a platform built to closely mimic Twitter's technical design, show that similar architectures can foster very different conversational structures when embedded in different communities (Cinelli et al. 2021).

Truth Social and Bluesky provide a natural comparison for studying how candidate-related political conversations unfold in ideologically distinct online environments. Rather than emphasizing only how far or how fast information travels, our analysis focuses on the shapes of cascades: whether content spreads broadly through direct amplification or develops sequentially through extended exchanges.

Building on this insight, we compare the structural diffusion of political content on Truth Social and Bluesky as a case comparison during a high-salience period of the 2024 U.S. presidential campaign. The analysis centers on a one-month window that included major events such as the first presidential debate, focusing on posts containing hashtags and keywords related to "Biden" and "Trump." This setting captures how political exchanges unfold at a moment of peak public attention, when participation is both intense and ideologically charged.

We ask: Which aspects of diffusion are consistent across platforms, and which vary with community composition and interaction norms?

Theoretical Framework and Hypotheses

Defining Diffusion Modes and Scaling

To address our research question, we distinguish between two modes of diffusion. Reposts capture endorsement behaviors, signals of agreement and amplification that spread content outward. Replies, by contrast, capture conversational engagement, where users respond, contest, or extend discussions. This distinction allows us to separate what is broadly universal in online diffusion (patterns of endorsement) from what is platform-specific (the conversational architectures that emerge in reply threads).

Rather than emphasizing only how far or how fast information travels, our analysis focuses on the shapes of cascades: whether content spreads broadly through direct amplification or develops sequentially through extended exchanges. Prior work has shown that depth, breadth, and structural virality all correlate positively with cascade size (Juul and Ugander 2021). This raises a concern: marginal comparisons may conflate structural effects with sheer volume, making it difficult to disentangle mechanisms of growth. To address this concern, we hold the cascade size constant and examine structural differences directly, analyzing how cascades grow vertically (depth) and horizontally (breadth).

Motivated by observed differences in reply cascade geometry across platforms, we propose three candidate mechanisms that may account for this divergence. These explanations draw from prior work in network science, computational social science, and political communication, and span different levels of influence: user prominence, content semantics, and ideological composition and alignment.

H1: Influencer Dominance

Prior research has demonstrated that a small number of high-follower users can disproportionately influence content dissemination, particularly on platforms characterized by skewed follower distributions or centralized influence structures (Goel et al. 2016; Zannettou et al. 2018). On Twitter/X, across 55 million posts and 520 million reposts, content reshared by high-influence accounts is more likely to be reshared again, and a small elite of accounts carries roughly half of all repost-cascade flow (Niitsuma et al. 2025). Truth Social, with its relatively small and ideologically concentrated user base, offers a fertile environment for such dynamics. This is further supported by the power-law distributions observed in the number of followers, reposts, and replies per user, as detailed in the Appendix. Overall, Truth Social exhibits a more pronounced heavy-tailed distribution compared to Bluesky, with a handful of dominant users, primarily right-leaning, occupying central roles in terms of follower count. To identify these influential users, we examined the follower count distribution and observed a clear discontinuity separating the top eight users from the rest, providing a natural cutoff for defining unusually high influence (see Appendix). We hypothesize that this concentration of influence contributes to reply cascades that are broad in reach but shallow in depth, indicative of immediate visibility rather than sustained conversational engagement.

H2: Ideological Structure

The ideology of users in a cascade, both in terms of who participates and how they interact, can shape its structure. Ideologically homogeneous groups often generate more uniform and shallow discussions, whereas cross-cutting exchanges promote greater depth, branching, and contention (Conover et al. 2011; Garimella et al. 2018; Barberá et al. 2015; Adamic and Glance 2005; Guess et al. 2023).

We measure the role of ideology in shaping cascade structure along two dimensions: (1) the ideological composition of users and (2) the alignment of their interactions. The ideological composition is defined as the proportion of users from each ideology within a reply tree, capturing the dominant perspective of a conversation. Alignment is measured as the share of reply edges connecting users of the same ideology; lower alignment indicates more ideologically heterogeneous interactions.

We hypothesize that ideological alignment and user composition are the key factors shaping cascade structure. Specifically, cascades with more ideologically diverse participants and lower alignment ratios are expected to develop deeper and more layered patterns, particularly on platforms such as Bluesky, where cross-cutting discussion is more viable. Our models build on this logic, illustrating why incorporating these factors into the models substantially attenuates platform-level differences.

H3: Topic Variability

The content of a post also plays a crucial role in shaping how users interact with it. Content that is emotionally charged, controversial, or conspiratorial often spreads through deeper and more structurally viral cascades than neutral or factual information (Vosoughi, Roy, and Aral 2018; Wang et al. 2022). In this study, we identified eleven salient topics across both platforms. Although all topics are politically related, the distribution of emphasis varies substantially between Truth Social and Bluesky. The procedure for topic identification, evaluation (with results validated through human judgment), and assignment is detailed in Appendix.

We hypothesize that certain topics, particularly those involving partisan conflict, institutional distrust, or election-related narratives, may elicit distinct structural diffusion patterns depending on their resonance within each platform's community. For example, election-related topics (e.g., presidential debates) are more prevalent on certain platforms. If such topics also diffuse differently, these differences may help explain the contrasting cascade structures we observe.

Micro-Level Network Motifs

To further unpack the mechanisms underlying platform-level differences in reply cascade geometry, we move from cascade-level regressions to a finer-grained motif analysis.

While our statistical models evaluate whether alignment ratio and ideological composition control divergence, they do not reveal *how* these compositional factors shape the micro-dynamics of interaction. Motif analysis, a graph-theoretic approach that examines the frequency of small recurring substructures within larger networks (Alon 2007), offers precisely this lens.

We analyze two basic 3-node motif shapes: *chains*, where replies unfold in sequence, and *stars*, where multiple replies attach to the same root. Each motif is labeled by the ideological stance of its participants: Left (L), Center (C), or Right (R).

Data and Methods

Data

We collected posts and their associated reposts and replies from both platforms during a critical month of the 2024 U.S. presidential campaign, when major political events (including the first debate) spurred intense online discussion. While our sampling window is limited, it offers a revealing snapshot of how these communities process high-salience political content.

We used each platform's public API to collect posts containing the target keywords and their surrounding conversations. For Bluesky, the thread endpoint enabled retrieval of entire discussion trees around posts mentioning "Biden" or "Trump," while for Truth Social (which follows the Mastodon API structure), we used the context endpoint to capture both ancestor and descendant replies recursively. This approach provides full conversational context rather than isolated messages, allowing us to reconstruct entire reply cascades. For every post and reply, we also collected associated repost actions and each user's follower list, allowing us to link diffusion patterns to the underlying network of connections. Data were collected between May 30 and June 30, 2024, a high-salience period that included the first U.S. presidential debate. Other collections covering Truth Social across the same campaign are available and use comparable endpoints (Shah et al. 2024). Cascade counts and size distributions for both platforms and both cascade types are reported in Appendix Table 2.

Modeling Structural Differences in Information Cascades

Baseline Model. To rigorously assess how cascade structure varies across platforms, we model the scaling relationship between cascade size and breadth using robust regression with the Huber loss function. This approach mitigates the influence of outliers, common in heavy-tailed cascade data, and addresses heteroskedasticity in residuals (see Appendix for additional justification).

Our baseline model examines how cascade breadth and depth scale with size, conditional on platform. Specifically, we include a platform indicator and its interaction with log-transformed cascade size:

$$\begin{aligned}\log(Y_i) = \beta_0 &+ \beta_1 \log(\text{size}_i)\\ &+ \beta_2 \cdot \text{platform}_i\\ &+ \beta_3 \cdot (\log(\text{size}_i) \times \text{platform}_i) + \varepsilon_i, \end{aligned} \tag{1}$$

where Y_i denotes either cascade breadth or cascade depth. The coefficient of primary interest is the interaction term, β_3, which captures whether the relationship between cascade size and breadth differs by platform. A significant β_3 implies that the slope of the log–log relationship varies systematically across platforms.

To quantify the explanatory contribution of these additional covariates, we track how the platform-by-size interaction changes between models. Specifically, we compute the percent attenuation of β_3:

$$\Delta = 1 - \frac{\hat{\beta}_{3,\text{aug}}}{\hat{\beta}_{3,\text{base}}}, \tag{2}$$

where $\hat{\beta}_{3,\text{base}}$ denotes the interaction coefficient from Eq. 1, and $\hat{\beta}_{3,\text{aug}}$ the corresponding estimate obtained from the augmented specifications (ie., Models 2–4). A larger Δ indicates that the added covariates account for a greater portion of the cross-platform difference.

By comparing baseline and augmented models, we thus directly assess whether observed platform-specific scaling patterns persist once structural and author-level features are taken into account.

Incorporating Influencer Dynamics. To evaluate the *Influencer Dominance Hypothesis*, we refine the platform variable to distinguish influencer-initiated cascades on Truth Social. In the baseline model, the platform variable platform_i takes two values: 0 for Bluesky and 1 for Truth Social. We extend this variable to include a third category: 2 for Truth Social cascades initiated by high-follower users, while 1 for Truth Social cascades initiated by others. Such a refinement is only necessary for Truth Social, because the platform is uniquely shaped by a small set of highly dominant accounts, most prominently Trump himself and seven other users with exceptionally large followings. No comparable concentration of outlier accounts exists on Bluesky, where user influence is more evenly distributed.

The revised model retains the same functional form as Equation 1, with platform_i now representing three cascade types. This formulation enables us to compare non-influencer cascades in TruthSocial against cascades from Bluesky.

Assessing Ideological Composition and Alignment Effects. We first assign each user an ideological label. Labels are inferred from aggregate posting behavior using the Llama-3.1-8B-Instruct model and validated against human annotations ($\kappa = 0.61$ and 0.74 against the two annotators, with class-specific and platform-specific performance and bootstrap intervals in Appendix Table 3, and an analysis showing that the results reported below are unchanged when the labels are randomly corrupted at the error rates we measured). Because large language models may encode biases, we treat these labels as probabilistic indicators rather than ground truth. Each user's ideology is inferred from the proportion of their posts identified as left-, center-, or right-leaning, and we tested different cutoff values (e.g., classifying a user as left-leaning if more than 60% of their posts are labeled as left). The aggregate patterns we report remain consistent across these threshold choices, indicating that our results are not sensitive to any particular cutoff (the sensitivity analysis can be seen in Appendix).

Building on these labels, we define interaction across ideological lines. Each reply edge is coded as aligned if the two users share the same stance, and a cascade's alignment ratio is defined as the proportion of aligned edges. A worked example makes the quantity concrete. Consider a thread with four replies: a right-leaning user replies to a right-leaning root author, a second right-leaning user replies to the same root, a left-leaning user replies to the root, and that left-leaning comment draws a reply from a right-leaning user. Three of these four edges connect users who share a stance and one crosses stances, so the alignment ratio is $3/4 = 0.75$. A ratio of 1 describes a conversation in which nobody replies across ideological lines; a ratio near 0 describes one in which almost every reply does. The measure is a property of the conversation rather than of any participant: the same user contributes aligned edges in one thread and crossing edges in another. To capture nonlinear effects, particularly at extremes of homogeneity, we apply a spline transformation (df = 3) and interact this term with cascade size and platform indicators (see Appendix). This measure reflects the extent of cross-cutting interaction, following prior work that links ideological heterogeneity to conversational complexity (Adamic and Glance 2005; Guess et al. 2023).

We then measure the ideological composition of each reply tree. For each reply tree, we compute the proportion of replies authored by left-leaning and right-leaning users, using the centrist share as the reference category. The unit here is the reply, while the label is a property of its author, so a user who replies repeatedly weighs proportionally more in a cascade's composition than one who replies once. To account for potential nonlinearities, we tested spline specifications for both shares. Diagnostics indicated that a spline (df = 3) better captures the effect of the left-user ratio, while the right-user ratio is sufficiently modeled linearly; full details are reported in Appendix. These measures capture the degree of ideological homogeneity or imbalance, which prior work has connected to polarization and engagement dynamics (Conover et al. 2011; Garimella et al. 2018; Barberá et al. 2015). As a robustness check, we also test a majority/minority specification, where the majority corresponds to the platform-dominant ideology (right-leaning on Truth Social, left-leaning on Bluesky) and the minority corresponds to the opposing ideology. This formulation compares majority versus minority status within each platform, abstracting away from the specific left/right distributional asymmetries across platforms. While this alternative reproduces the main pattern, its incremental improvement is limited. We therefore retain the left/right specification as our primary model, reporting results from the majority/minority formulation in Appendix.

Taken together, these measures distinguish between who participates in a conversation (ideological composition) and how those users interact (alignment). By incorporating both into the regression framework, we assess whether structural divergence in reply cascades is better explained by participant makeup, communication patterns, or their interaction.

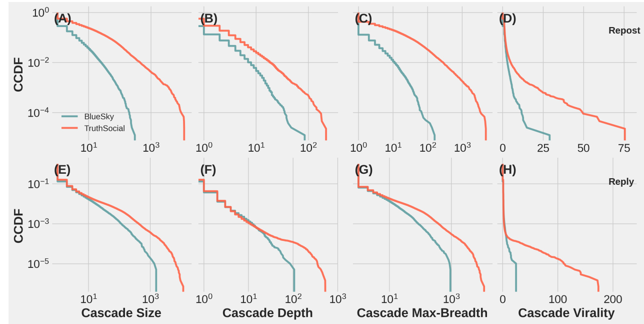

Figure 1: (A–D) Complementary cumulative distribution functions (CCDFs) of repost cascade statistics across two platforms: BlueSky and TruthSocial). Each subplot represents a different structural property: Size, Depth, Breadth, and Structural Virality (Goel et al. 2016). (E–H) The same analyses applied to reply cascades on BlueSky and TruthSocial using the same metrics. X-axes are on a log scale (except for structural virality), and y-axes are plotted on a log scale throughout. Each line represents a platform's empirical distribution. Tick marks are retained for alignment, and rows are annotated to distinguish reposts from replies.

Evaluating Topic Variability. To address the *Topic Variability Hypothesis*, we include cascade-level topic annotations as categorical moderators. Topics are interacted with $\log(\text{size}_i)$ and platform_i to test whether differences in cascade content account for cross-platform variation in how depth and breadth scale with size. The main text describes our core modeling procedures, including the regression framework and mechanisms evaluated. Additional methodological details are provided in Appendix, including full regression outputs, justification for identifying outlier users, diagnostics for model assumptions, and supplementary data analyses such as follower distribution patterns and agreement validation.

Results

Diffusion Structure Across Emerging Social Networks

As shown in Figure 1, cascades on Truth Social appear larger, deeper, and broader than those on Bluesky for both reposts and replies, and they also show higher levels of structural virality. These descriptive comparisons raise a natural question: what explains the apparent discrepancy in cascade shape across platforms? A straightforward possibility is platform size, since larger user bases mechanically allow cascades to expand further. Prior work has shown that depth, breadth, and structural virality all correlate positively with cascade size(Juul and Ugander 2021). This raises a concern: marginal comparisons may conflate structural effects with

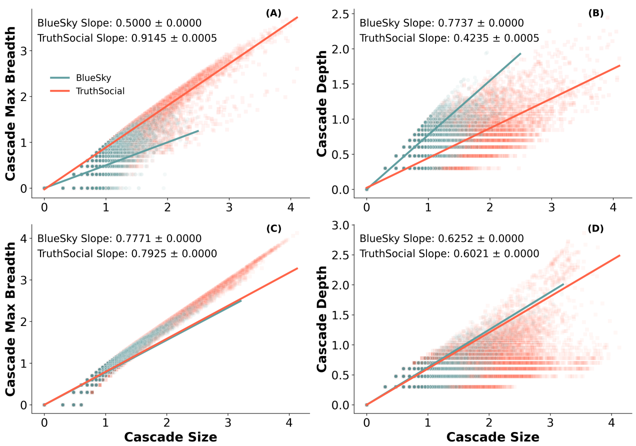

Figure 2: (A–B) Robust Linear Models (RLM) regressions of log-transformed cascade breadth (A) and depth (B) on log-transformed cascade size for reply cascades on BlueSky and TruthSocial. (C–D) Same regressions for repost cascades. Point clouds represent individual cascades (semi-transparent), and lines represent RLM fits. Blue denotes BlueSky, and red denotes TruthSocial. Slope coefficients and their standard errors are annotated in each subplot. Cascade depth is computed as $\log(\text{depth}+1)$ to avoid undefined values for shallow trees. All axes are on a logarithmic scale. Each panel uses matched styling to support cross-platform comparisons.

sheer volume, making it difficult to disentangle mechanisms of growth. To address this concern, we hold the cascade size constant and examine structural differences directly. Specifically, we analyze how cascades grow along two dimensions: vertically (depth) and horizontally (breadth). Variation in these scaling slopes reveals how user engagement patterns shift as discussions unfold.

Figure 2 presents robust linear fits between log-transformed cascade size and both depth and breadth. After conditioning on cascade size, differences in repost behavior largely disappear: repost cascades scale similarly across Truth Social and Bluesky, with nearly identical slopes. In contrast, reply cascades exhibit a marked divergence. On Truth Social, replies tend to fan out, producing broad but shallow trees, whereas on Bluesky, conversations deepen through sequential exchange, forming narrower but more layered structures. This asymmetry suggests that while baseline amplification behaviors (i.e., reposts) are consistent, patterns of dialogic engagement diverge. Because repost cascades are reconstructed rather than observed, we verified that this similarity is not an artifact of the reconstruction: under every alternative linking rule, including a 40-draw random-rule ensemble, the repost platform-by-size interaction stays within ± 0.083, while the reply interaction is larger than the most extreme repost estimate by a factor of four for breadth and two for depth (Appendix, Figure 5).

Mechanisms of Structural Divergence

The modeling results, reported in Table 1, highlight clear differences in explanatory power across the three mechanisms. Model 2 (*Influencers*) indicates that cascades initiated by high-follower accounts on Truth Social are only marginally broader on average, and the platform–size slope gap remains largely unchanged relative to the baseline. This suggests that the broad–shallow structure of Truth Social cascades is not simply the product of a small elite of prominent accounts, but rather reflects more widespread interaction patterns across the platform.

Model 3 (*Ideological Structure*) is decomposed into three specifications. Model 3a incorporates only the alignment ratio, Model 3b incorporates only user-level ideological composition, and Model 3c combines both. This decomposition shows that composition alone (Model 3b) reduces the divergence, yet substantial residual differences persist (0.289 for breadth and -0.196 for depth). Alignment alone (Model 3a) likewise attenuates but does not eliminate platform differences: the size-platform interaction for breadth remains positive (0.236), and the size-platform interaction for depth remains negative (-0.189). By contrast, the combined model (3c) reduces both interaction terms by roughly five-sixths relative to the baseline, to 0.071 for breadth and -0.049 for depth against baseline values of 0.419 and -0.321. Only when both ideological composition and cross-ideological alignment are included together does the platform-specific scaling gap shrink this far, though for breadth a small positive interaction remains.

Model fit statistics reinforce this conclusion. While Models 3a and 3b improve pseudo-R^2 and RMSE relative to the baseline, the gains are modest. The combined specification produces the strongest fit, with pseudo-R^2 increasing to 0.95 for breadth and 0.91 for depth, with corresponding reductions in RMSE, indicating that ideological structure jointly accounts for a large share of the observed divergence. Substantively, the platform difference is associated with both within-cascade conflict (alignment) and broader differences in the ideological makeup of user communities, and neither covariate absorbs it alone.

Model 4 (*Topic Variability*) shows no such effect. The platform-by-size interaction is essentially unchanged from the baseline once topic is included (0.426 against 0.419 for breadth, -0.330 against -0.321 for depth), indicating that differences in what the two platforms discuss do not account for the difference in how their cascades grow.

In sum, the augmented models make clear that structural differences in reply cascades are not well accounted for by the presence of influential users or topics alone. They are instead closely associated with the ideological composition of participating users and the alignment of their interactions: including these factors absorbs most of the platform-specific divergence in scaling, so that the platform difference and these ideological measures are largely coextensive in these data.

One limit on how far this result can be pushed deserves emphasis. The alignment ratio is computed from the same reply edges whose arrangement produces the cascade's depth and breadth, so it is not a variable measured prior to

Table 1: Interaction Effects of Size (Log) and Platform on Breadth (Log) and Depth (Log)

Model	Size (Log)	Size (Log) × TruthSocial	*pseudo* R^2	RMSE
Breadth (Log)				
Model 1 (Baseline)	0.5043 (0.0010)	+0.4187 (0.0012)	0.9176	0.1319
Model 2 (Influencers)	0.5042 (0.0010)	+0.4121 (0.0012)†	0.9182	0.1314
Model 3a (Alignment)	0.1713 (0.0012)	+0.2362 (0.0007)	0.9432	0.1095
Model 3b (Composition)	0.3692 (0.0014)	+0.2887 (0.0019)	0.9364	0.1159
Model 3c (Combined)	0.2137 (0.0016)	+0.0706 (0.0015)	0.9546	0.0979
Model 4 (Topic)	0.5140 (0.0014)	+0.4260 (0.0012)	0.9191	0.1307
Depth (Log)				
Model 1 (Baseline)	0.7320 (0.0008)	−0.3214 (0.0009)	0.8608	0.1031
Model 2 (Influencers)	0.7322 (0.0008)	−0.3134 (0.0009)†	0.8618	0.1027
Model 3a (Alignment)	0.9097 (0.0010)	−0.1889 (0.0006)	0.8926	0.0905
Model 3b (Composition)	0.8432 (0.0011)	−0.1955 (0.0015)	0.8870	0.0929
Model 3c (Combined)	0.9497 (0.0014)	−0.0491 (0.0013)	0.9085	0.0835
Model 4 (Topic)	0.7168 (0.0011)	−0.3302 (0.0009)	0.8623	0.1025

Notes: Robust linear models (Huber M-estimator) fitted with Huber's proposal 2 scale estimate; asymptotic standard errors in parentheses. Every coefficient shown is more than 25 standard errors from zero, so significance stars are omitted as uninformative. $N = 123{,}189$ for every model. The second column reports the interaction between $\log(\text{size})$ and the platform indicator.

† Model 2 replaces the binary platform indicator with a three-category variable (Bluesky, Truth Social non-influencer, Truth Social influencer); the entry shown is the Truth Social non-influencer contrast against Bluesky.

the cascade and then observed to shape it. Influence may run in either direction, and plausibly runs in both: a deep, chain-like exchange offers repeated opportunities for a reply to cross ideological lines, while a wide, star-like cascade of parallel replies to a single root offers few. Ideological composition is less exposed to this concern, since who participates is determined by an audience that exists before the conversation, but it is still measured on the realized set of participants. We therefore report these as associations that are consistent with a mediating role for ideological structure, and we do not claim that changing alignment would change cascade shape. A design capable of separating the directions, such as one predicting a cascade's eventual shape from the ideological composition of its earliest replies, is left to future work.

Motif-level interaction patterns

Figure 3 shows how often three-node chain motifs occur relative to randomized baselines. We organize motifs into four types: fully aligned sequences, mid-shift transitions (where the middle reply differs ideologically), end-shift transitions (where the final reply differs), and all-different triads.

Fully aligned Center chains are consistently more frequent than expected on both platforms, indicating that center users often talk with one another. In contrast, fully aligned Right chains show no over- or under-representation on either platform: they are common in absolute terms on Truth Social, but no more so than the ideological composition of the platform already implies. Left-leaning chains diverge across platforms: Bluesky shows an overrepresentation, whereas Truth Social shows an underrepresentation.

The strongest contrasts appear in mid-shift "bounce-back" motifs. On Bluesky, all bounce-back chain motifs are overrepresented, indicating that cross-cutting replies across ideological pairings routinely draw immediate responses. On Truth Social, by contrast, only Left↔Right bounce-backs are overrepresented, but at much higher magnitudes than on Bluesky. This suggests that Bluesky fosters a broad pattern of routine cross-cutting engagement, whereas Truth Social channels such interaction almost entirely into intense confrontations between opposing extremes.

By contrast, end-shift transitions (e.g., C–C–R, L–L–C, L–C–C), which place a cross-cutting reply at the conversational boundary, do not follow a uniform pattern across platforms. On Bluesky, end-shifts are generally at or below the rewired baseline, with L↔C links most underrepresented. On Truth Social, however, several center-involved end-shift motifs (R-C-C, R-R-L, L-C-C) are significantly overrepresented, suggesting that while the dominant bloc rarely incorporates cross-cutting voices, boundary exchanges among moderates, or between moderates and rare left-leaning participants, occur more often than expected by chance. This indicates that, in Truth Social, while the dominant bloc rarely incorporates cross-cutting voices, minority (i.e., Center and Left) users sometimes establish local footholds at the conversational boundary without being fully integrated into mainstream exchanges.

For star motifs, as shown in Figure 4, platform differences are more selective than for chains but remain revealing. On Truth Social, fully aligned C–C–C and R–R–R stars are strongly overrepresented, while on Bluesky, no comparable over- or under-representation appears. In contrast, L–L–L stars are overrepresented on Bluesky but underrepresented on Truth Social. Truth Social also shows several mixed configurations (L–R–R, L–C–C, C–R–R, R–L–L) as overrepresented, indicating that diverse voices often cluster around a single root. These bursts of root-focused replies inflate breadth relative to depth: where chain motifs highlighted

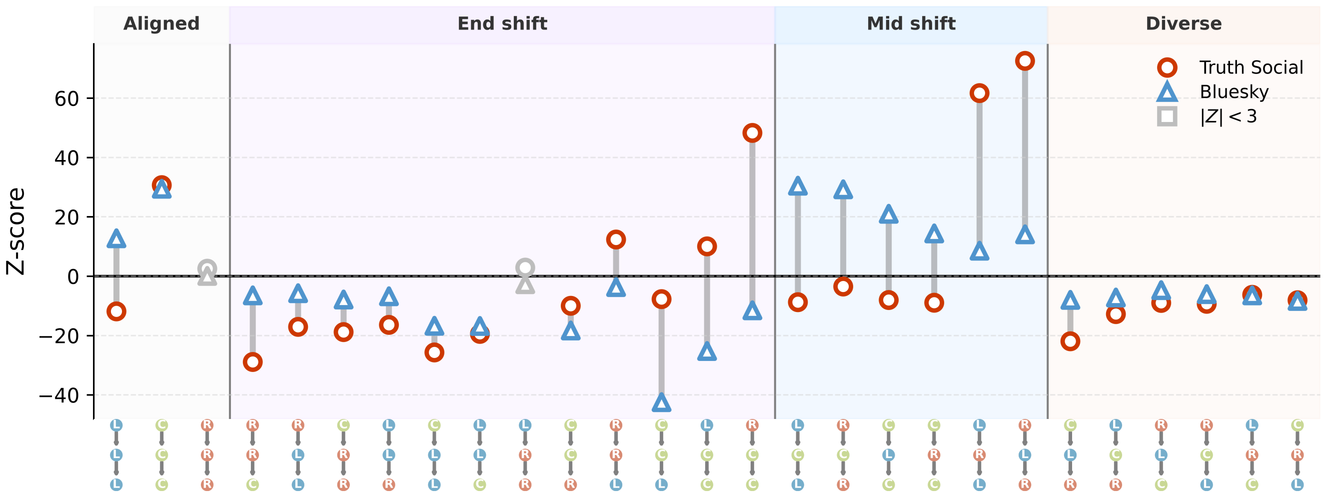

Figure 3: **Chain motif analysis of cross-platform reply structures.** Standardized differences (Z-scores) in the frequency of three-node chain motifs between observed cascades and a randomized null model. Node colors mark ideology (Left = blue, Center = green, Right = red); background shading groups motifs as aligned, mid-shift (middle node differs), end-shift (first or last node differs), or all-different. Circles represent Truth Social, triangles Bluesky; gray symbols denote non-significant values ($|Z| < 3$). Center-aligned chains are overrepresented on both platforms, while Bluesky shows broad enrichment of cross-cutting "bounce-backs" and Truth Social concentrates them in intense Left–Right confrontations. Motifs are grouped into four families, named in the band above the panel. The families differ in how many of a chain's two edges cross ideological lines: none in an aligned chain, one in an end-shift chain, and both in mid-shift and all-different chains. Z-scores measure over- or under-representation against the null model rather than absolute frequency; observed counts for every motif are given in Table 4.

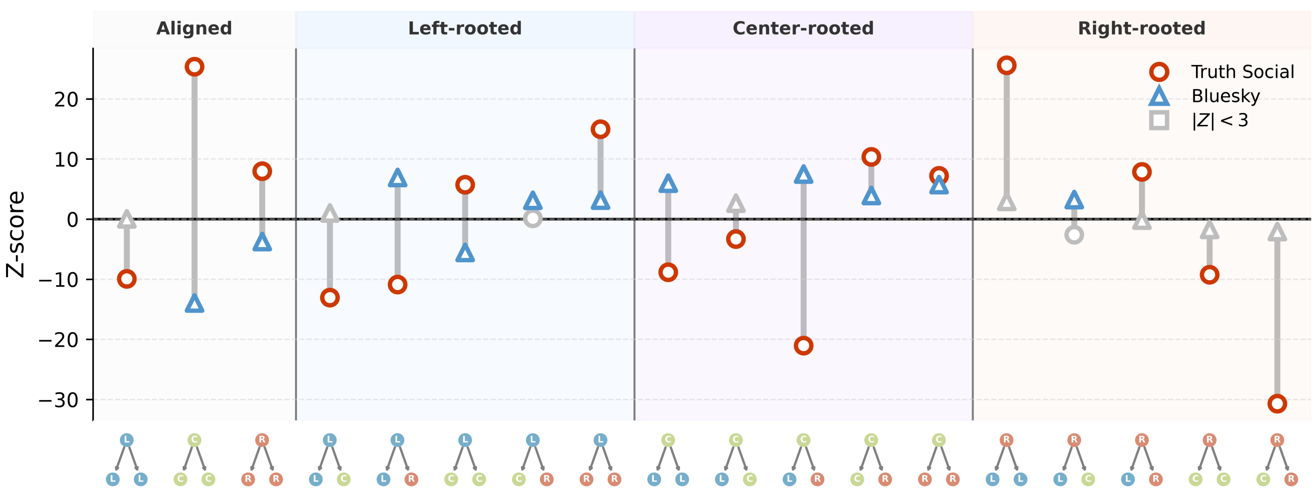

Figure 4: **Star motif analysis.** Standardized Z-scores of three-node star motif frequencies relative to degree-preserving null models. Circles mark Truth Social, triangles Bluesky; gray symbols indicate non-significant values ($|Z| < 3$). Motifs are grouped in the band above the panel: fully aligned stars first, then mixed stars by the ideology of the root, which is the account whose post drew the two replies. Symmetric leaf pairs are merged, so each column is one distinct configuration. Truth Social shows strong overrepresentation of aligned C–C–C and R–R–R stars and several mixed forms (e.g., L–R–R, L–C–C), indicating that diverse replies cluster around a single root. Bluesky lacks these concentrations, underscoring its deeper, more distributed conversational patterns. As in Figure 3, z-scores indicate over- or under-representation relative to the null model rather than absolute frequency; observed counts appear in Table 4. The null model is described under Motif Frequencies and the Null Model.

bounce-back exchanges, star motifs show Truth Social's tendency to channel heterogeneous engagement into short, centralized reply structures. While their interpretive weight is weaker than sequential chains, star motifs reinforce the broader picture that Truth Social organizes cross-cutting interactions into shallow, initiator-focused bursts rather than sustained dialogue.

Overall, motif analysis corroborates the regression results: once ideological composition and alignment are controlled, platform-level differences in reply cascade geometry largely disappear, and motif patterns illuminate the micro-level pathways behind this convergence. Bounce-back motifs are overrepresented on both platforms, but with different profiles: on Truth Social they concentrate in direct Left–Right confrontations, while on Bluesky, they span a wider range of pairings. End-shift motifs are not uniformly suppressed; instead, they follow partisan logics. On Bluesky, Left–Center end-shifts are sharply underrepresented, whereas on Truth Social, several boundary configurations (R–C–C, R–R–L, L–C–C) are overrepresented, indicating that cross-cutting uptake occurs mainly at the margins (i.e., Center and Left users) rather than through incorporation by the dominant bloc (i.e., Right users). Star motifs reinforce this picture: Truth Social shows strong overrepresentation of aligned Center and Right star motifs, along with several mixed forms, channeling diverse voices into short, root-focused bursts instead of extended exchanges. Together, these patterns highlight the micro-mechanisms of alignment: extremes confront one another through bounce-backs, centers rarely sustain end-shift handoffs, and heterogeneous stars concentrate activity at the root. In this way, motif analysis demonstrates how the alignment effects that absorb platform differences in regression models are themselves grounded in systematic differences in local conversational structure.

Discussion

This study compares how candidate-centered conversations about the 2024 U.S. presidential election unfold on two ideologically distinct but technically comparable platforms, Truth Social and Bluesky. Whereas most prior work emphasizes how far or how fast information spreads, we focus on its structural form: whether interactions broaden outward through shallow branching or deepen through sequential layering. By analyzing both reposts and replies, we show that reposts scale in similar ways across platforms, but reply cascades diverge sharply: Truth Social supports broader, shallower exchanges, while Bluesky produces narrower, deeper threads. These results demonstrate that identical affordances can give rise to fundamentally different conversational dynamics, depending on the social and ideological composition of the communities that use them.

Our modeling results show that ideological composition and alignment together statistically account for most of the platform divergence: cross-ideological exchanges are associated with greater depth, whereas more ideologically homogeneous cascades are associated with flatter structures. Yet alignment alone is insufficient. Even when disagreement occurs, reply structures on Truth Social often remain shallow. Motif analysis clarifies this pattern: on Truth Social, disagreement frequently appears as star motifs, parallel replies directed at a single post. On Bluesky, by contrast, disagreement more often takes the form of chain motifs, where replies build sequentially and support multi-turn interaction.

Motif results highlight the combined role of interactional structure and user composition. On Truth Social, left-leaning users constitute about a tenth of the population, yet when present, they appear disproportionately in Left↔Right bounce-back motifs. This produces a paradox: such confrontational motifs are strongly overrepresented relative to the null model, but their absolute frequency is low, yielding the broad and shallow cascades seen in aggregate. In other words, bounce-backs mark where cross-ideological debate occurs, but the scarcity of left-leaning users limits their influence on cascade structure. This also explains why alignment alone cannot account for platform differences: it captures whether interactions are within- or across-ideology, but not the underlying distribution of ideological groups. Composition must be considered to capture the rarity of certain user types.

These patterns also demonstrate something that prior studies have largely overlooked: the structural role of Center users. On Bluesky, they often appear as interlocutors in bounce-back motifs, serving as the targets of cross-ideological replies and sustaining reciprocal exchanges. On Truth Social, by contrast, they surface more frequently in end-shift motifs, acting as boundary bridges that connect minority voices without being integrated into dominant cascades. This contrast shows why centrist voices are simultaneously visible in local motifs yet remain marginal in aggregate cascade structures.

The distinction between wide, shallow and narrow, deep cascades also has implications for how political discussion is organized. Wide, shallow reply cascades are characterized by many users responding directly or near-directly to the same post, with relatively little subsequent exchange among participants. Narrower, deeper cascades involve more sequential interaction, in which replies themselves receive replies and discussions extend across multiple turns. Neither structure should be interpreted as inherently more desirable. Rather, they represent different forms of conversational organization: one is more concentrated around responses to the original post, whereas the other involves more sustained exchange across successive replies. That these patterns differ across platforms with similar reply mechanics suggests that comparable technical affordances do not necessarily produce comparable conversational structures.

This distinction also has a methodological implication for cross-platform research. Comparisons based only on aggregate activity or cascade size would primarily capture the higher level of activity on Truth Social and would miss differences in how cascades develop conditional on size. Examining structural measures such as breadth and depth therefore provides information about conversational organization that is not captured by volume alone. More generally, our results suggest that the composition of the participating community should be considered alongside platform affordances when comparing interaction patterns across platforms.

Although Truth Social and Bluesky share nearly identical technical affordances, their user bases differ in size and composition. These contextual differences likely interact with diffusion in ways our models cannot fully capture. We therefore interpret the observed patterns not as direct effects of design, but as emergent properties of user behavior within distinct sociotechnical contexts. Beyond shared reply and repost mechanics, latent factors such as algorithmic filtering (which remains opaque) and, more importantly, self-selection into ideologically distinct communities shape how conversations unfold. Prior work on migration to Bluesky further suggests that platform adoption was associated with users' existing social connections and prior political expression (Quelle et al. 2025), reinforcing the importance of self-selection when interpreting differences in community composition. Both platforms exhibit partisan clustering, but the communicative style and discursive norms associated with being "conservative" or "liberal" may differ markedly across sites. Who joins, who stays, and what forms of interaction are socially rewarded together offer a more plausible explanation for the structural divergence we observe than affordances alone.

While our analysis focuses on observable conversation structures, unobserved mechanisms such as moderation practices and algorithmic ranking may also contribute to the patterns we observe.

Both platforms expose users to content through partially opaque visibility pathways: on Bluesky, chronological feeds and decentralized moderation may amplify sustained exchanges, whereas on Truth Social, algorithmic curation and selective content removal could favor brief, attention-grabbing interactions.

Such processes would interact with user composition; the same affordances may produce different outcomes when visibility and moderation differentially reward certain kinds of engagement.

The motif analysis, which shows distinct frequencies of cross-ideological and centrist interactions, is consistent with this interpretation: the configurations we observe could reflect not only who participates but which exchanges are made more visible within each system.

These findings highlight a subtler dimension of online political discourse: not only what circulates or who engages, but how conversations are organized. On one platform, disagreement tends to manifest as expression, parallel replies directed outward; on the other, as interaction, reciprocal replies unfolding over time. These distinct conversational structures carry implications for deliberative potential. When disagreement remains structurally shallow, exposure to opposing views may produce fleeting reactions without meaningful exchange. By contrast, deeper cascades may provide opportunities for contestation, clarification, or understanding, though not necessarily consensus.

Several limitations merit consideration. First, our analysis focuses on a single month during a politically salient period; whether these patterns hold across time and issues remains to be tested. More specifically, our data cover conversations mentioning Biden or Trump on two U.S.-centered platforms during one month of the 2024 presidential campaign, a period of unusually high political attention. This setting places several bounds on interpretation. First, the absolute levels of cascade depth and breadth may be specific to this period. Second, the ideological composition observed in our data reflects both the campaign context and our candidate-centered sampling frame and should not be interpreted as representing the full user population of either platform. Third, our analysis does not establish that the same structural patterns extend to other political topics or to non-political discussion. The more limited conclusion supported by our analysis is that platforms with similar reply mechanics can exhibit systematically different conversational structures conditional on cascade size, and that these differences are associated with who participates and how participants interact across ideological lines. Establishing how broadly this pattern generalizes will require comparable analyses across other platforms, countries, time periods, and topics. We therefore regard the present study as a case comparison rather than evidence of a general relationship across social media environments.

Second, our measure of ideological alignment is coarse, relying on a simplified typology that may not capture subtler dimensions of identity. Third, while we account for cascade size, content, and user prominence, unobserved platform dynamics, such as visibility algorithms or moderation policies, may also shape interaction. Finally, we use "technical similarity" in a narrow sense. Both Truth Social and Bluesky provide Twitter-like affordances for posting, reposting, and threaded replying, which allows us to construct comparable repost and reply cascades across platforms. This comparison is also supported by their software and protocol lineages: Bluesky is built on the open-source AT Protocol ecosystem, while Truth Social has been widely documented as a Mastodon-derived platform. These similarities make the observed interaction traces structurally comparable for cascade analysis.

However, this does not imply that the platforms are identical in feed ranking, moderation, recommendation, or thread presentation. Bluesky, for example, supports algorithmic and user-selectable feeds through the AT Protocol feed-generator architecture, while Truth Social may implement platform-specific visibility and moderation rules on top of its Mastodon-derived base. We therefore treat technical similarity as a condition for constructing comparable cascade objects, not as an assumption that platform design is fully held constant.

Nonetheless, by combining cascade-level modeling with motif analysis, this study advances a structural perspective on political diffusion. Rather than attributing outcomes to platform design alone, we describe emergent interaction patterns shaped by the communities and contexts in which they unfold. Understanding online political discourse thus requires not only tracing who speaks or what spreads, but also examining how conversational structures scaffold engagement itself.

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(a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)?
Yes, subject to platform terms and privacy constraints.

(b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)?
Not applicable. We use an existing instruction-tuned model and do not train a new model.

(c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)?
Not applicable for model training. Where estimates carry sampling uncertainty we report it: percentile bootstrap intervals on every validation rate, distributions over 100 simulation draws for the label-noise analysis, and confidence intervals on every regression coefficient reported in the robustness analyses.

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Not applicable.

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Yes.

Cascade Construction via Time-Inferred Diffusion

To investigate information diffusion dynamics on *Truth Social* and *Bluesky*, we constructed cascade networks for both **reposts** and **replies**, drawing inspiration from the time, inferred diffusion method introduced by Vosoughi et al(Vosoughi, Roy, and Aral 2018). Each cascade is a directed acyclic graph in which each node is a single post: the root post, a reply, or a repost. Nodes are posts rather than accounts, so a user who replies several times in the same thread contributes one node per reply; how this choice enters each structural metric is stated under Cascade Metrics below. Ideology, by contrast, is aggregated at the user level: each node carries the label of the account that wrote it, assigned by thresholding that account's posts (see Ideology Estimation and Aggregation). A cascade's ideological composition is therefore the share of its posts written by left- or right-leaning accounts, and a user who replies several times contributes to that share several times. The same holds for alignment, which compares the labels of the two accounts at either end of a reply edge. Edges represent inferred diffusion pathways.

Repost Cascades.

Because the platform APIs do not provide the true repost paths (i.e., who reshared from whom), all reposts are nominally linked to the original post. However, users may often repost another user's repost rather than the original. To approximate the true repost cascade, we reconstructed repost trees using the time-ordered repost list together with a post-collection snapshot of follower-followee relationships.

Specifically, for each repost r_i, we identified all prior reposts r_j such that $t_j < t_i$, and among those, filtered for users that r_i is observed to follow in the follower snapshot.

We then inferred the edge $r_j \rightarrow r_i$, linking each repost to the most recent prior repost by a user the reposting user follows. If no such user existed, the repost was attributed directly to the original post. This procedure infers the most probable diffusion paths based on temporal and social proximity. Because these edges are inferred rather than observed, we refer to repost cascades as reconstructed throughout, and we verify that every repost-cascade result is invariant to the reconstruction rule (see Robustness of Repost Cascade Reconstruction below). DeVerna et al. (2024) show, for Twitter and Bluesky cascades, that assumptions made during reconstruction can distort both local and global properties of the resulting networks, and that no ground truth exists against which a reconstruction can be checked directly. Our response is to establish which conclusions hold across the space of admissible reconstructions rather than to defend one of them. The follower network is a post-collection snapshot; the implications of follow edges that may postdate a repost are discussed there as well.

Reply Cascades.

Reply trees were constructed directly from reply metadata, linking each reply to its parent post. As with reposts, each reply was treated as a node, and an edge was created from the parent post to the reply. Cascades were rooted at the original post, and subsequent replies were linked recursively based on reply chains.

Cascade Metrics.

To characterize the structure of repost and reply cascades, we computed four core metrics: **size**, **breadth**, **depth**, and **structural virality**. These measures provide complementary views of cascade growth, reach, and shape, and are adapted from prior studies of information diffusion (Vosoughi, Roy, and Aral 2018; Goel et al. 2016).

- **Size** (N): The total number of posts in the cascade, including the root. For repost cascades, N is the number of reposts plus the root post; because a user rarely reposts the same post twice, this is in practice the number of distinct reposting users plus one. For reply cascades, N is the number of posts in the thread, so a user who replies several times contributes each reply to N.
- **Breadth** (B): The maximum number of nodes appearing at any depth level in the cascade tree (i.e., maximum breadth). Formally, let $n(d)$ denote the number of nodes at depth d (with the root at $d = 0$); then
$$B = \max_{d \geq 1} n(d).$$
This definition captures the widest "layer" of the cascade and differs from *root out-degree*, which counts only direct children of the root.
- **Depth** (D): The maximum distance (in hops) from the root node to any leaf node in the cascade tree. Formally,
$$D = \max_{v} \mathrm{dist}(\mathrm{root}, v),$$
where $\mathrm{dist}(\cdot, \cdot)$ is the shortest-path distance along tree edges. In repost cascades, depth is computed on the observed repost tree when parent reposts are available, or on the time-inferred tree when parents are inferred; in reply cascades, depth is computed on the reply tree.
- **Structural Virality**: The average pairwise shortest-path distance between all nodes in the cascade. Formally, for a cascade with n nodes, virality is defined as:
$$v = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1}^{n} d_{ij}$$
where d_{ij} is the shortest path length between nodes i and j. Higher values indicate more distributed (viral-like) cascades, while lower values indicate shallow, broadcast-like structures.

All metrics were computed on the final time-inferred cascade DAGs, and were log-transformed for visualization and modeling due to heavy-tailed distributions.

				Share of cascades by size (%)			
Type	Platform	Cascades	Posts	1	2–5	6–50	>50
Reply	Bluesky	79,348	199,413	71.3	21.2	7.2	0.3
Reply	Truth Soc.	43,841	1,369,497	43.3	25.2	22.5	9.0
Repost	Bluesky	25,550	184,807	46.3	33.6	17.8	2.3
Repost	Truth Soc.	218,579	3,847,903	51.4	29.7	13.9	5.0

Table 2: Cascade counts and size distributions. For reply cascades, size is the number of posts in the thread including the root, and Posts sums these nodes. For repost cascades, size here is the number of reposts (the root adds one node in the analyses), and Posts counts reposts. Reply cascades of size 1 are root posts that received no replies.

Descriptive Statistics

Table 2 reports cascade counts, post counts, and the cascade size distribution for both platforms and both cascade types.

Robustness of Repost Cascade Reconstruction

Any reconstruction rule could in principle shape the structural comparison (DeVerna et al. 2024), so we re-derived every repost cascade under alternative rules and re-estimated the platform-by-size interaction (β_3 in Eq. 1) for each. Every admissible rule selects each repost's parent from the same candidate set: the original post, plus earlier reposters whom the reposting user follows. We compare the rule used in the paper against the earliest eligible reposter, the most recent eligible reposter, and a uniformly random eligible reposter (40 independent draws), each evaluated under both readings of the repost list order, since the platform APIs return reposts in time order but do not document its direction.

Figure 5 summarizes the result over 244,129 repost cascades (4.03 million reposts). No rule yields a breadth interaction outside $[-0.036, +0.038]$ or a depth interaction outside $[-0.083, +0.028]$, and the random-rule ensemble concentrates near zero. The same estimator applied to reply cascades gives $+0.170$ (breadth) and -0.182 (depth), exceeding the most extreme repost estimate under any rule by a

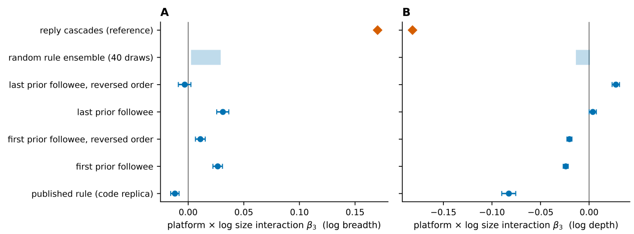

Figure 5: Platform-by-size interaction β_3 for repost cascades under alternative reconstruction rules (dots with 95% confidence intervals; shaded band, range over 40 random-rule draws), for log breadth (A) and log depth (B). The diamond marks the reply-cascade interaction under the same estimator. All reconstruction rules leave repost scaling nearly identical across platforms.

factor of four and two respectively. The rule used in the paper is the most conservative case: every alternative moves the repost estimates closer to zero. The cross-platform similarity of repost scaling therefore cannot be an artifact of the reconstruction procedure. (Because log breadth and log depth of small trees take few distinct values, the scale estimate of a Huber robust regression degenerates on repost cascades; the interactions in this analysis are estimated by OLS with HC3 standard errors, with a Huber fit under a non-degenerate scale estimator as a check.)

The rule has limited room to matter in the first place: 55% of Bluesky reposts and 59% of Truth Social reposts have no eligible prior reposter and attach to the root under every rule, and a further 21% and 12% have exactly one candidate. A second concern is that the follower network is a post-collection snapshot without edge creation dates, so a follow created after a repost could produce a spurious parent; such an error can only move a repost from the root to an interior parent, a displacement already spanned by the rule set above.

Deleting follow edges at random at rates from 5% to 30% to simulate such edges moves the breadth interaction only from 0.031 to 0.005 and the depth interaction from 0.004 to 0.013, both far below the reply values. This error need not be symmetric across platforms, since the two follow networks were collected at different times relative to the posting window.

Power-Law Analysis of User-Level Activity Distributions

We analyzed the distribution of user-level activity, specifically the number of followers, reposts, and replies per user, across both platforms using power-law fits. Figure 6 shows the complementary cumulative distribution functions (CCDFs) of these metrics in log-log scale, overlaid with fitted power-law tails (dashed orange lines). For each distribution, we report the estimated power-law exponent γ and the lower cutoff $x_{\min}$ above which the power-law holds.

The distribution of follower counts per user reveals strong skewness on both platforms. On Truth Social, a small number of users amass an exceptionally large number of followers, with a power-law exponent $\gamma = 1.76$ and $x_{\min} \approx$ 13,515. In contrast, BlueSky users show a slightly less extreme concentration, with $\gamma = 1.84$ and $x_{\min} \approx 1{,}498$.

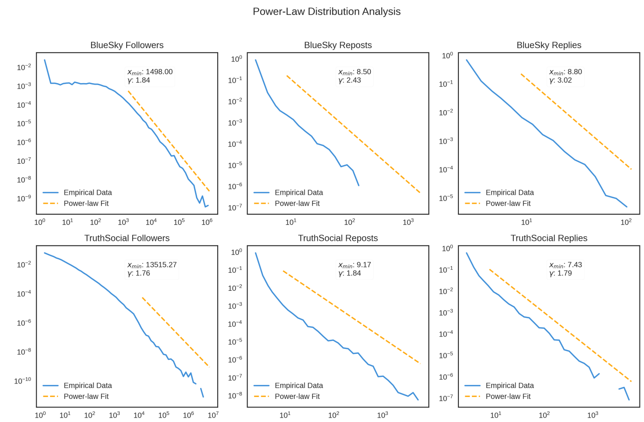

Figure 6: Follower count distributions across platforms.

Repost and reply behaviors also exhibit heavy-tailed patterns. On BlueSky, the average number of replies per user decays rapidly ($\gamma = 3.02$), while reposts follow a shallower slope ($\gamma = 2.43$), suggesting higher variability in repost activity. Truth Social users display flatter distributions for both reposts ($\gamma = 1.84$) and replies ($\gamma = 1.79$), consistent with broader participation from users in redistributing and responding to content.

Exclusion of Highly Followed Users.

To identify unusually influential users, we examined the follower-count distribution of every account in our data (134,100 on Truth Social; 41,566 on Bluesky). Both distributions are heavy tailed (Figure 6), but their upper tails differ in kind, not merely in degree. On Truth Social, the top eight accounts are separated from the body of the distribution: the largest account (Donald Trump, with 7,107,831 followers) has 56 times the followers of the platform's 99.9th-percentile account (127,082), the eighth-largest account still holds 2,170,535 followers, and together these eight hold 16.4% of the total follower count in our data. On Bluesky, no comparable class exists: the largest account (1,230,128 followers) sits five times above the 99.9th percentile (238,048), continuous with the rest of the distribution, and the top eight accounts hold 6.1% of the total. Notably, the eighth-largest Truth Social account exceeds the largest Bluesky account.

We therefore designate the eight separated Truth Social accounts as influencers in the refined platform variable and apply no exclusion on Bluesky. A symmetric rule, such as excluding the top n percent of accounts on both platforms, would not equalize the comparison: on Bluesky it would remove accounts that are continuous with the rest of the distribution, while on Truth Social any n large enough to capture the separated class would also sweep in ordinary accounts. The asymmetric treatment reflects an asymmetry in the data.

Topic Modeling

To identify dominant themes in social media discourse, we applied unsupervised topic modeling using the `BERTopic` framework. This method integrates contextual sentence embeddings, dimensionality reduction, clustering, and class-based TF-IDF to generate semantically coherent topic representations.

Each post was first embedded using the `all-mpnet-base-v2` sentence transformer model, producing a 768-dimensional vector. We then applied UMAP (Uniform Manifold Approximation and Projection) to reduce the embedding dimensionality and used HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) to group posts into dense semantic clusters.

Topics were extracted from each cluster using class-based TF-IDF over a preprocessed vocabulary. We used `CountVectorizer` with a minimum document frequency of 10 and a maximum document frequency of 0.9 to filter out rare and overly common terms. For each topic, the top-ranked n-grams based on c-TF-IDF were used to construct candidate topic labels.

To ensure topic interpretability, two human annotators independently reviewed a sample of 10 representative posts from each cluster. Topics were finalized through consensus-based manual labeling, resulting in 11 coherent and interpretable topic categories. Posts not confidently assigned to any cluster were labeled with a default topic ID of `-1` and excluded from topic-based analyses.

We assessed the alignment between human and model-assigned topic labels by computing Cohen's κ coefficient. Inter-annotator agreement between the two human experts was moderate ($\kappa = 0.53$). Agreement between the model and Human 1 yielded $\kappa = 0.67$, while agreement between the model and Human 2 was $\kappa = 0.67$, suggesting that the model-generated clusters were generally consistent with human interpretations of topic structure.

To compare topical emphasis across platforms, we plotted the distribution of root posts by topic label for *Bluesky* and *Truth Social* (Fig. 7). While both platforms exhibited substantial engagement with topics surrounding Donald Trump's legal issues and the 2024 U.S. presidential election, there were notable differences. Posts on *Bluesky* were more likely to focus on legal accountability, international affairs (e.g., the Israel–Hamas conflict), and broader campaign discourse. In contrast, *Truth Social* featured a larger proportion of posts celebrating Trump (e.g., birthdays and tributes) and promoting pro-Trump narratives (e.g., "MAGA advocacy"), reflecting differences in user base ideology and agenda-setting priorities. These topical variations underscore the divergent discourse ecosystems that have emerged across platforms.

Ideology Estimation and Aggregation

Post-Level Stance Detection.

To measure political alignment within reply cascades, we employed a large language model to perform stance detection on individual replies. Specifically, we used the `Llama-3.1-8B-Instruct` model, a state-of-the-art instruction-tuned model, to classify each reply into one of five ideological stances: *left*, *lean-left*, *center*, *lean-right*, or *right*. To account for the dialogic and context-sensitive nature of replies, each input included the original post and preceding replies when available.

For interpretability and robustness, we grouped stances into three categories: *left-leaning* (left, lean-left), *center*, and *right-leaning* (lean-right, right). Each reply node in the cascade was labeled with its ideological stance accordingly.

To further contextualize alignment patterns across platforms, we examined the ideological distribution of users who authored root posts (i.e., the original posts that initiate reply cascades). As shown in Figure 8, *Bluesky* exhibited a predominantly left-leaning user base, with over 46,000 root posts originating from users labeled as left-leaning and 27,495 from right-leaning users. Centrists contributed a comparatively small number of posts (5,853). Conversely, *Truth Social* was overwhelmingly right-leaning, with 37,639 right-authored root posts versus just 5,149 from left-leaning users and 1,053 from centrists.

These aggregate distributions highlight the platform-level asymmetry in ideological representation and reinforce the need to aggregate stance labels at the user level for modeling ideological composition and alignment in reply cascades. We describe this aggregation procedure in the next section.

User-Level Ideology Inference via Threshold Aggregation.

While reply-level stance labels capture the ideological tone of individual posts, many of our analyses required consistent user-level ideological classifications. To infer user ideology, we aggregated stance labels across all posts authored by each user. Specifically, we computed the proportion of posts labeled as left-leaning or right-leaning.

A user was classified as *left-leaning* if more than 60% of their posts were labeled as left or lean-left, and as *right-leaning* if more than 60% were labeled as right or lean-right. Users not exceeding the 60% threshold for either pole were labeled as *centrist*.

This 60% threshold was chosen to balance ideological signal with label consistency. Many high-volume users produce a mix of political and apolitical content (e.g., greetings or non-informative messages), and a more lenient threshold (e.g., 50%) would risk misclassification due to incidental variation. The 60% threshold ensures a stronger signal of consistent ideology while retaining sufficient sample size.

Our threshold choice is also supported by prior behavioral research. A Pew Research Center study found that Democrats consistently expressed liberal views around 60% of the time, with Republicans showing comparable consistency in expressing conservative views[1]. This provides a behavioral benchmark for interpreting ideology as a probabilistic but consistent trait.

[1]Pew Research Center. (2014). Political Polarization in the American Public. https://www.pewresearch.org/politics/2014/06/12/political-polarization-in-the-american-public/

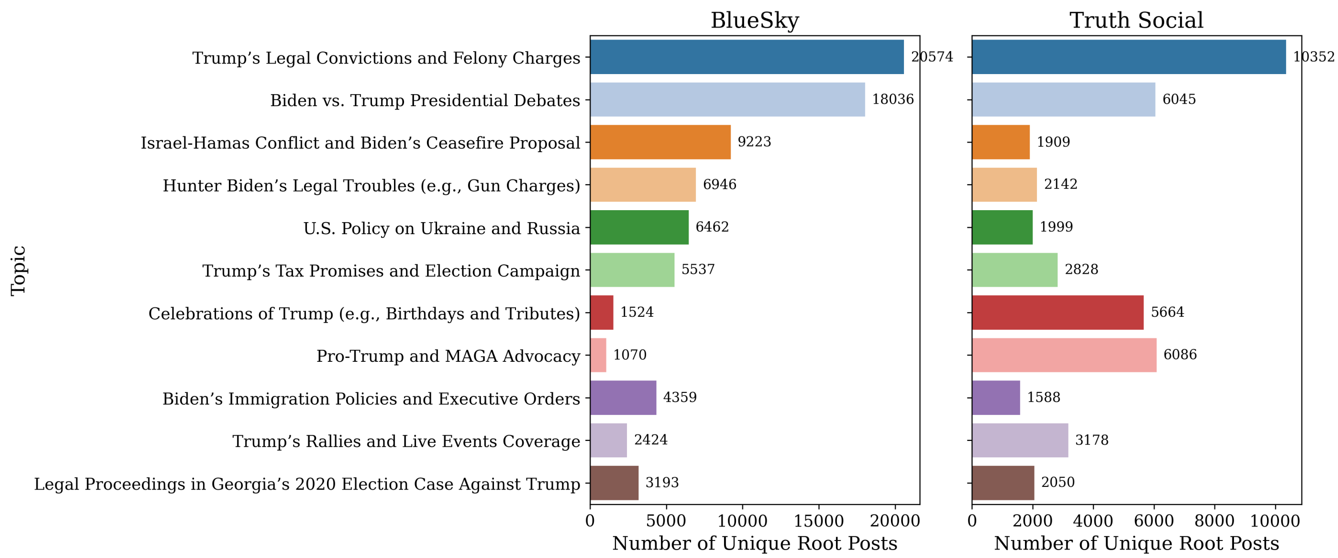

Figure 7: Topic distribution across *Bluesky* and *Truth Social*. Bar plots show the number of unique root posts assigned to each of the 11 topics, based on BERTopic clustering and manual labeling. While both platforms heavily feature discussions about Trump's legal cases and the upcoming U.S. presidential election, *Bluesky* content is more concentrated around policy and legal accountability, whereas *Truth Social* emphasizes celebratory or supportive content related to Trump. These platform-level differences in topic salience inform subsequent analyses of cascade structure and alignment.

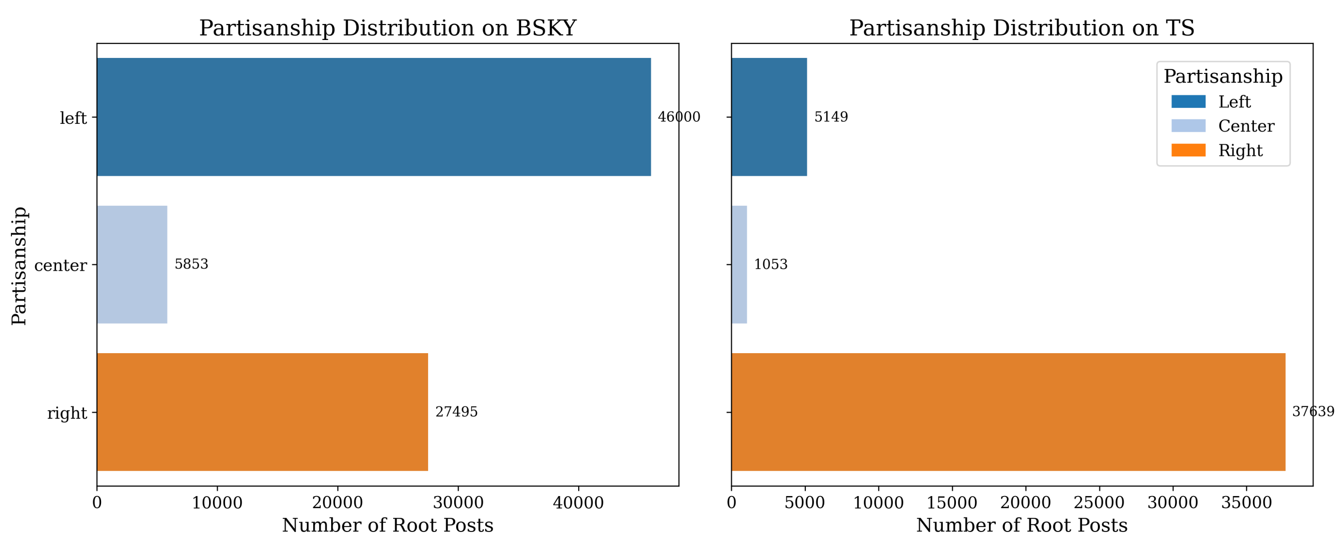

Figure 8: Distribution of root post partisanship across platforms. Bar plots show the number of root posts authored by users with left-leaning, centrist, or right-leaning stance labels, as detected by the LLM. On *Bluesky*, left-leaning users dominate root-post activity, followed by right-leaning and centrist users. In contrast, *Truth Social* exhibits a pronounced right-leaning skew, with over 37,000 root posts authored by right-leaning users and fewer than 6,500 from left or centrist users combined. These distributions reflect the ideological clustering of user bases on each platform and provide context for interpreting downstream alignment patterns.

To assess robustness, we examined how user ideology distributions and average label consistency varied across thresholds from 0.5 to 0.7. As shown in Figure 9, center-leaning users increase with higher thresholds, while average consistency remains stable across the threshold range. This suggests that the classification is robust to moderate changes in thresholding criteria.

We also applied an unsupervised natural breakpoint detection algorithm to each platform's ideology distribution, which identified 0.67 as a statistically distinct split point. However, to avoid excessive sparsity in user categories, we adopt 0.6 as our default threshold and conduct robustness checks using 0.5 and 0.7 as alternative specifications.

Alignment Ratio Based on User Ideology.

To quantify ideological cohesion within reply cascades, we compute an **alignment ratio** based on user-level ideological labels. For each directed reply edge, we compared the ideological categories of the replying user and the user they responded to. An edge was labeled as **aligned** if both users shared the same ideology (left-leaning, center, or right-leaning), and **misaligned** otherwise.

The alignment ratio for each cascade was then defined as the proportion of aligned reply edges:

$$\text{Alignment Ratio} = \frac{\text{\# Aligned Edges}}{\text{Total \# of Edges}}$$

This edge-based measure captures the extent of ideological homophily within conversational structures. High alignment ratios indicate ideologically homogeneous discussions, while lower ratios reflect cross-cutting or ideologically heterogeneous conversations. This measure forms the basis for testing whether ideological diversity within cascades moderates structural depth or breadth across platforms.

Validation of LLM Stance Annotations.

To assess the reliability of stance annotations, two authors independently labeled a random sample of 200 replies from the dataset, 86 from Bluesky and 114 from Truth Social. The model's five-category output (left, lean left, center, lean right, right) was collapsed to left, center, and right for all analyses, and the validation uses the same collapsed scheme; the "center" user category is a residual, covering users whose posts reach neither the left nor the right threshold. Annotators were blinded to model predictions and to each other's labels.

Inter-annotator agreement is substantial (Cohen's $\kappa = 0.77$, bootstrap 95% interval $[0.70, 0.85]$), and model–human agreement is moderate by the conventional bands (Cohen's $\kappa = 0.61$ with Annotator 1 and 0.74 with Annotator 2). Table 3 reports class-specific and platform-specific precision and recall against the 171 items on which both annotators agree, with bootstrap intervals so that the precision of each rate is visible. Overall accuracy against consensus is 0.86.

The errors are not uniform across classes, and the pattern matters for interpreting our composition estimates: the model over-assigns each platform's ideological minority.

Scope	Class	Precision [95% CI]	Recall	n
Pooled	left	0.68 [0.54, 0.82]	0.80	35
Pooled	center	0.96 [0.90, 1.00]	0.85	59
Pooled	right	0.89 [0.81, 0.95]	0.90	77
Bluesky	left	0.74 [0.53, 0.93]	0.78	18
Bluesky	center	0.98 [0.92, 1.00]	0.89	45
Bluesky	right	0.64 [0.33, 0.91]	0.88	8
Truth Social	left	0.64 [0.42, 0.83]	0.82	17
Truth Social	center	0.91 [0.69, 1.00]	0.71	14
Truth Social	right	0.93 [0.86, 0.98]	0.90	69

Table 3: Class-specific and platform-specific model performance against human consensus labels (171 of 200 validation items on which both annotators agree). Intervals are percentile bootstrap 95% intervals over 2,000 resamples of the validation items; n is the consensus support per class. The two minority cells, right on Bluesky and left on Truth Social, carry the least support and the widest intervals.

Precision is 0.64 for right-leaning labels on Bluesky and 0.64 for left-leaning labels on Truth Social, against 0.98 and 0.93 for the largest class on each platform (center on Bluesky, right on Truth Social). These two minority cells rest on 8 and 17 consensus items respectively, and their bootstrap intervals are correspondingly wide (for right on Bluesky, $[0.33, 0.91]$). Uneven performance across the ideological spectrum is a documented property of language-model annotators rather than a quirk of our pipeline: Vallejo Vera and Driggers (2025) find that annotation reliability varies substantially with the political position being labeled, so the weakest cell being each platform's minority is the pattern one would expect. We therefore treat the direction of this bias as established and its magnitude as imprecisely estimated, and the label-noise analysis below tests whether the substantive conclusions survive errors at these rates.

Most disagreements involved ambiguous, sarcastic, or low-information posts, reflecting common challenges in short-form political discourse.

How Far the Validation Sample Can Be Pushed.

Two hundred labeled items support some statements more securely than others, and the bootstrap intervals in Table 3 make the difference visible: the pooled rates and both majority classes are tightly estimated, while right on Bluesky rests on 8 consensus items with an interval running from 0.33 to 0.91.

First, the claim we actually make from this table is directional: the classifier over-assigns each platform's ideological minority. That claim can be tested even where individual cells are imprecise, by bootstrapping the difference between minority-class and majority-class precision within each platform. The difference is -0.34 on Bluesky (95% interval $[-0.65, -0.06]$, one-sided $p = 0.017$) and -0.29 on Truth Social ($[-0.51, -0.08]$, $p = 0.004$). The minority classes are measurably worse on both platforms, even though we cannot pin either rate to a narrow interval.

Second, the reported rates are computed against the 171

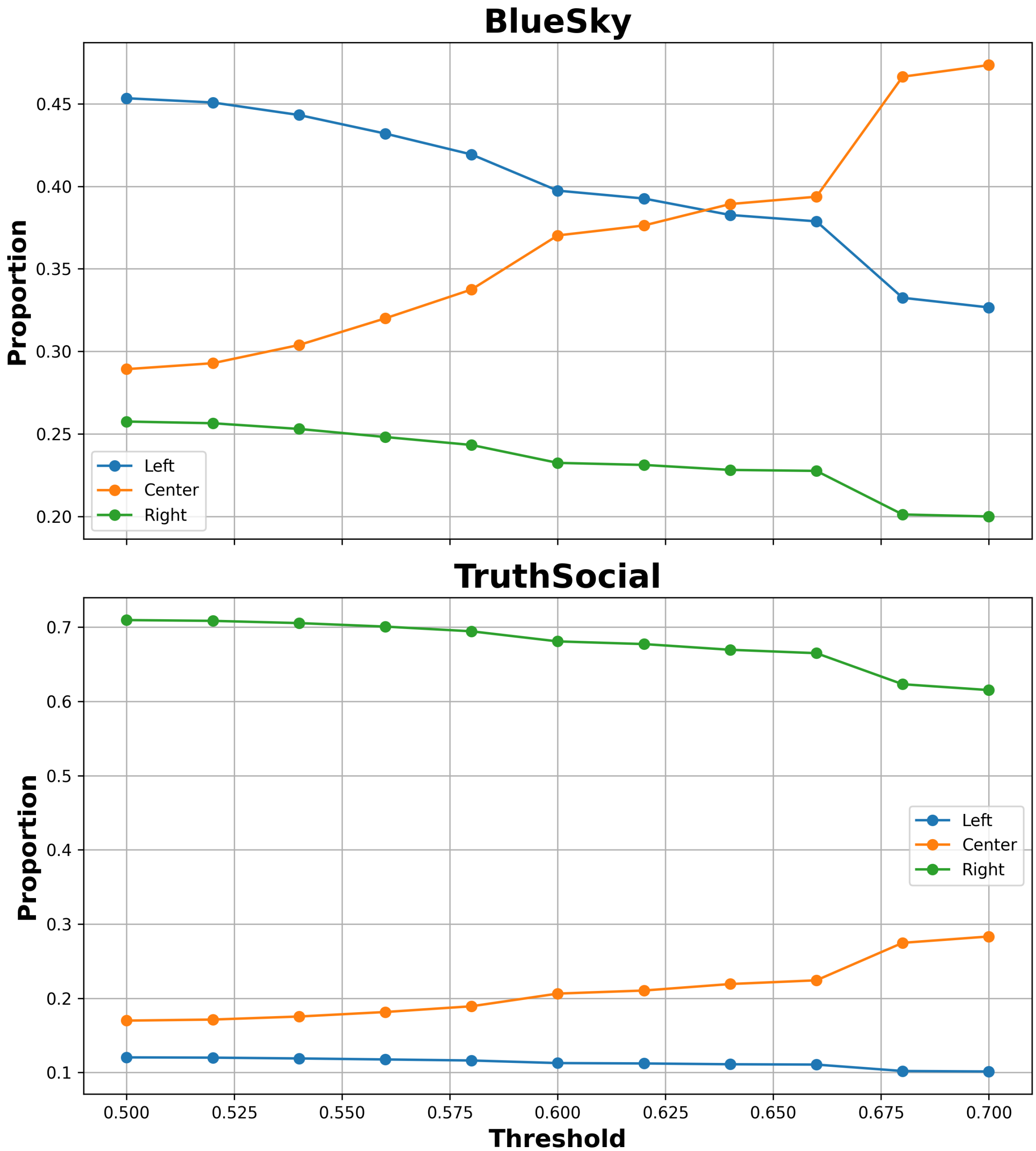

Figure 9: User ideology proportions and consistency across thresholds on *Bluesky* and *Truth Social*. For *Bluesky*, Left-leaning users dominate at lower thresholds, while center-leaning users increase in proportion as the threshold tightens. For *Truth Social*, Right-leaning users represent the majority at all thresholds. The rise of center-labeled users at higher thresholds mirrors the Bluesky pattern.

items where both annotators agree, and consensus items are by construction the ones annotators found easy, which flatters the classifier. Recomputing against each annotator alone gives lower precision throughout, most sharply for right on Bluesky (0.33 against Annotator 1, 0.57 against Annotator 2, 0.64 against consensus), but leaves the ordering unchanged: on both platforms and under all three references, the minority class has the lowest precision and the majority class the highest. The qualitative pattern does not depend on which reference is used. Because a single-annotator reference is the less favorable one, we use it below as a worst case.

Third, the validation measures accuracy at the post level, whereas the analysis uses user-level labels formed by thresholding each user's posts. Simulating the threshold rule with the measured post-level confusion and the observed posts-per-user distribution, accuracy rises from 0.85 to 0.88 on Bluesky and from 0.81 to 0.87 on Truth Social. The gain is modest, since the median user in our sample contributes two posts, but it runs in the direction of making the noise analysis below conservative: we perturb user labels at post-level rates, which overstates the error the pipeline actually carries.

Robustness to Ideology Label Noise.

Because the stance labels contain measurement error, we tested whether the central mechanism result (Model 3c) survives label noise at the measured rates. Ignoring such error is not safe by default: misclassification by automated classifiers biases downstream regression even when overall accuracy is high (TeBlunthuis, Hase, and Chan 2024), and the same holds for language-model labels used as surrogates in social-science inference (Egami et al. 2023). Those authors correct the bias analytically with a gold-standard subsample. We take the complementary route of propagating the measured error through the full pipeline, because our labels enter not as a single regressor but through composition, alignment, splines, and interactions, where an analytic correction is not available. For each of 100 simulation draws, every user's label was replaced by a draw from the column-normalized confusion distribution $P(\text{human} \mid \text{model})$ estimated from Table 3; each cascade's ideological composition and alignment ratio were then recomputed from the perturbed labels, and Model 3c was refit. We ran the simulation under four error processes: the measured platform-specific rates; a pooled matrix that applies each platform's dominant class the error rate measured largely on the other platform; a nested bootstrap that resamples the validation items and rebuilds the confusion matrix on each draw, so that the uncertainty in Table 3 is carried through; and the harsher rates measured against a single annotator, under which precision for right-leaning labels on Bluesky falls to 0.33. Relative to the baseline interactions without ideology controls (0.398 and -0.334), the median draw absorbs between 86% and 96% of the platform divergence, and the least favorable of the four hundred draws absorbs 70%. No error process we simulated restores the platform gap. All fits here share one scale estimate and compare perturbed against unperturbed values, so the intervals are ranges across simulation draws rather than standard errors. Under every misclassification process we simulated, adding label noise failed to restore the platform gap.

Ideology triple	Star: Bluesky	Star: Truth Soc.	Chain: Bluesky	Chain: Truth Soc.
L–L–L	13,436	1,423,162	6,519	4,747
L–L–C	20,482	1,361,062	2,659	3,270
L–L–R	3,869	4,721,187	346	11,883
L–C–L	18,970	1,695,796	4,966	3,727
L–C–C	31,468	2,929,973	3,113	7,194
L–C–R	6,179	6,432,294	290	8,869
L–R–L	3,751	4,837,988	471	26,476
L–R–C	5,846	5,384,674	191	10,673
L–R–R	1,462	20,194,975	88	40,264
C–L–L	16,997	158,383	1,533	2,040
C–L–C	36,051	137,961	6,197	3,390
C–L–R	8,058	437,959	249	5,929
C–C–L	30,243	142,810	2,930	3,421
C–C–C	74,502	197,087	25,971	13,261
C–C–R	15,559	458,039	980	8,179
C–R–L	6,786	466,118	161	7,371
C–R–C	15,173	458,308	1,518	8,093
C–R–R	4,093	1,929,621	171	16,393
R–L–L	407	11,197,554	89	15,565
R–L–C	841	6,961,901	111	14,127
R–L–R	281	24,241,501	307	66,065
R–C–L	778	8,183,288	111	15,549
R–C–C	1,735	6,435,618	384	30,188
R–C–R	625	19,234,403	819	48,366
R–R–L	238	23,352,704	74	45,060
R–R–C	448	15,651,387	163	30,149
R–R–R	224	64,855,781	357	128,058
Total	318,502	233,481,534	60,768	578,307

Table 4: Observed counts of ideology-labeled three-node motifs (L = left, C = center, R = right). For chains the triple is ordered along the reply path; for stars it lists the root followed by the two leaves.

Motif Frequencies and the Null Model

Table 4 reports the raw observed frequency of every ideology-labeled three-node motif on each platform, alongside which the z-scores in Figures 3 and 4 should be read. The two platforms differ by orders of magnitude in absolute motif volume, so a large z-score does not imply a common configuration: on Truth Social, for instance, Left↔Right chain motifs are strongly overrepresented relative to the null model while remaining rare in absolute terms compared with all-Right motifs. Reporting counts alongside z-scores keeps this distinction visible.

Motifs of this kind are not mutually independent: a chain of four nodes contains two overlapping three-node chains, so one structural feature can contribute several motif instances. This dependence does not bias the z-scores, because the null model is subjected to the identical counting procedure. We generate 100 randomized graphs by double-edge swaps performed within each cascade: two reply edges are chosen at random and their targets exchanged, together with the target's ideology label, with $5m$ swap attempts in a cascade of m edges. This procedure holds fixed the set of cascades and their sizes, each node's out-degree, the multiset of reply targets within a cascade, and the ideology attached to every participant, while randomizing who replies to whom. We then count motifs in each randomized graph with the same overlapping-instance enumeration used on the observed graph and compute the z-score of the observed count against that null distribution. Overlap therefore inflates observed and randomized counts alike, and the comparison remains like for like. What the dependence does affect is the joint interpretation of several motifs at once: because neighboring motif types share substructure, their z-scores are correlated, and we therefore read them as a pattern across families (aligned, mid-shift, end-shift, all-different) rather than treating each of the 54 values as independent evidence.

Topic Model Details

The topic model was fit over 125,623 root posts pooled across platforms. HDBSCAN initially assigns low-density documents to an outlier class; we applied BERTopic's outlier reduction step, which reassigns those documents to their nearest topic, leaving 21 documents (0.02%) unassigned. Clustering yields 12 topics plus a residual outlier class. Manual labeling then produced the 11 coherent and interpretable categories used in the analysis: the residual class, which retains only 21 posts after reduction, is excluded, and two near-duplicate pro-Trump categories are treated as one. Every cascade in the modeling sample carries one of these 11 labels. The largest are "Trump's Legal Convictions and Felony Charges" (31,305 posts), "Biden vs. Trump Presidential Debates" (24,539), and "Israel-Hamas Conflict and Biden's Ceasefire Proposal" (11,534). We note that outlier reduction trades coverage against precision: without it a substantial share of short posts would remain unassigned, and topic labels for reassigned documents are correspondingly less certain. Because topic enters our models only as a categorical moderator, and because the topic hypothesis is not supported (Model 4), this choice does not carry the paper's conclusions.

Two-User Exchanges in Reply Cascades

Depth would mean something different if it largely measured back-and-forth exchange between the same two accounts rather than extended multi-party discussion. We therefore counted, for every reply, whether its author also wrote its grandparent post with a different account in between, that is, an $A \to B \to A$ pattern. Among all cascades, 9.3% on Bluesky and 20.8% on Truth Social contain at least one such exchange, reflecting mainly that Truth Social cascades are larger. The comparison that matters conditions on cascades deep enough to permit the pattern at all: among cascades of depth two or more, the share containing a two-user exchange is 68.8% on Bluesky and 70.0% on Truth Social, and such exchanges account for 20.0% and 15.6% of reply edges respectively. Dyadic back-and-forth is therefore common on both platforms and to a very similar degree, so the depth difference we report is not explained by one platform hosting more two-user conversations than the other.

Sample	Scale est.	Outcome	Baseline	Model 3c	Absorbed
All	MAD	breadth	0.398	0.023	94%
All	MAD	depth	-0.334	-0.055	84%
All	Huber	breadth	0.419	0.071	83%
All	Huber	depth	-0.321	-0.049	85%
Size > 1	MAD	breadth	0.187	0.035	81%
Size > 1	MAD	depth	-0.223	0.019	91%
Size > 1	Huber	breadth	0.181	0.015	92%
Size > 1	Huber	depth	-0.217	0.029	87%

Table 5: Platform-by-size interaction before and after adding ideological composition and alignment, across samples and scale estimators. "Absorbed" is the proportion of the baseline interaction removed by Model 3c. $n = 123{,}189$ for all cascades and 47,670 for cascades with more than one post.

Sensitivity of the Modeling Specification

Two features of the reply-cascade data warrant an explicit sensitivity check. First, 61.3% of reply cascades consist of a root post with no replies (71.3% on Bluesky, 43.3% on Truth Social). These cascades are structurally degenerate: breadth and depth are fixed by definition, so they sit exactly at the origin of the log–log scaling plot and carry no information about how structure scales with size. Second, because this mass of exactly-fitted points makes the median absolute deviation of the residuals vanish, the default scale estimator of the robust regression collapses toward zero. Point estimates are unaffected, but standard errors computed from a near-zero scale are not interpretable.

Table 5 therefore re-estimates the baseline model and Model 3c across the four combinations of sample (all cascades, or only those with more than one post) and scale estimator (the default median absolute deviation, or Huber's proposal 2, which does not degenerate here). The platform-by-size interaction in the baseline model is smaller when single-post cascades are excluded, as expected once the degenerate points are removed, but the substantive result is stable throughout: including ideological composition and alignment absorbs between 81% and 94% of the baseline platform difference in every specification, for both breadth and depth. Table 1 reports the Huber proposal 2 fit on all cascades, the third and fourth rows here.

Unit of Analysis: Posts Versus Users

Reply cascades are trees of posts, so a user who replies several times in a thread appears once per reply. An alternative representation collapses each user to a single node, placing them at the shallowest depth at which they appear. Because the two representations can yield different depth and breadth for the same conversation, we recomputed all reply-cascade metrics both ways from the raw thread data. The post-level reconstruction reproduces the metrics used throughout the paper (correlations of 0.999 for size, 0.999 for breadth, and 0.988 for depth against the analysis file), confirming that the reported analyses are post-level.

Under the user-collapsed representation, the two platforms' depth scaling still diverges at baseline (-0.132, against -0.194 at post level) and is again largely absorbed by ideological composition and alignment. The breadth divergence, however, is much smaller under collapsing (-0.032, against 0.168 at post level). A substantial part of the raw breadth gap reflects the same Truth Social users posting repeatedly within a thread, which widens a level of the post tree without adding distinct participants. Conditional on size, Truth Social conversations are wider in posts, and part of that width comes from repeat participation rather than from a broader set of participants. The depth contrast, which is the finding the discussion rests on, holds under both representations.

The Center Category

The user-level "center" label is a residual: it collects users whose posts reach neither the left nor the right threshold, and therefore mixes genuinely moderate users, users who post about politics inconsistently, and users whose stance the classifier could not resolve. To test whether this residual is doing the explanatory work, we recomputed composition and alignment using partisan users only, so that center-labeled users enter as unlabeled rather than as a third ideological position; alignment is then measured over reply edges whose two endpoints are both partisan. On the resulting subset of 28,083 cascades, ideological composition and alignment still absorb 61% of the baseline breadth divergence and 64% of the depth divergence, compared with 99% and 77% under the published specification on the same sample. The conclusion is therefore weaker but not reversed when the center category is set aside. Part of the accounting in the three-category specification depends on separating users the classifier places confidently on one side from the residual center category. Because center collects moderate, inconsistent, and unresolved users alike, this contrast is not the same as one between partisan and non-partisan participation, and we cannot tell from these labels which of the three it mainly reflects.

Ideological Composition Compared With Prior Estimates

Our ideology labels place 41.4% of Bluesky users in our sample on the left, 34.4% at center, and 24.2% on the right; on Truth Social the shares are 11.5% left, 19.3% center, and 69.2% right. The Bluesky figures show a larger right-leaning presence than platform-wide estimates: Quelle and Bovet (2025) classify 75.3% of Bluesky users as left of center and only 4.8% as right of center, and Nogara et al. (2025), scoring active users by the political leaning of the domains they share, report 74.61% left-leaning, 18.17% centrist, and 7.21% right-leaning. Two independent platform-wide estimates therefore place the right-leaning share below 8%, well under our observed 24.2%. Two differences account for most of this gap, and we can quantify one of them.

First, the samples differ. Prior platform-wide estimates describe all users, most of whom post little political content, and are typically derived from the political leaning of the news domains a user shares. Our sample is restricted to accounts that posted in a conversation mentioning Biden or Trump during one month of a presidential campaign, and our labels come from the stance expressed in the post text itself. Later work on Bluesky reports a similarly homogeneous platform-wide composition (Salloum et al. 2025), which reinforces that the contrast is between platform populations and politically engaged conversation samples rather than between two measurements of the same quantity. A candidate-mention sample selects for politically engaged users on both sides, including users who arrive to argue with the platform's majority, so its ideological mix should be less lopsided than the platform's.

Second, part of the gap is measurement error, and our validation lets us estimate how much. The classifier over-assigns each platform's minority position (Table 3). Applying a standard prevalence correction, which inverts the measured confusion matrix to recover the label distribution implied by the observed one, lowers the estimated right-leaning share on Bluesky from 24.2% to 17.9% and the estimated left-leaning share on Truth Social from 11.5% to 1.9%. Both corrections move in the expected direction, and the correction on Truth Social is large. For Bluesky, however, a substantial gap with the platform-wide estimate remains after correction, so label error alone does not account for it; we attribute the remainder to the sampling frame, since a candidate-mention conversation sample selects for politically engaged users on both sides and in particular for those who arrive to contest the platform's majority view. These corrected values are indicative rather than precise, because the correction inherits the imprecision of the minority cells in Table 3. Our labels overstate each platform's ideological minority, our sample is not a platform-wide sample and should not be read as one, and the label-noise analysis above shows the structural findings survive errors of the measured size.

Modeling Strategy and Regression Framework

To quantify the explanatory power of our hypotheses and assess their ability to account for platform-specific variation in cascade structures, we modeled two key dependent variables: breadth and depth.

Initial Ordinary Least Squares (OLS) regression revealed strong signs of heteroskedasticity, consistent with the observation that the potential range of cascade breadth and depth expands with increasing cascade size. This structural constraint induces a fan-shaped pattern in the residuals, as confirmed by Figure 10. Furthermore, the residuals exhibited heavy tails, resembling a t-distribution, suggesting deviations from the normality assumption.

Although the Cook's Distance analysis identified a small number of observations with relatively higher influence, the vast majority fell well below the conventional threshold of $4/n$, indicating that no single data point exerted undue leverage on the model estimates. The heavy-tailed, heteroskedastic residuals motivate robust estimation, and all models reported in the main text are robust linear models with the Huber loss. The one analysis estimated by OLS with HC3 errors is the repost reconstruction robustness check, where lattice-valued outcomes make the Huber scale estimate degenerate (see Robustness of Repost Cascade Reconstruction).

Assessing the need for spline transformations

To examine potential non-linear effects of the predictors, we inspected partial residual plots for `author_left_ratio`, `author_right_ratio`, and `author_alignment_ratio` under a linear baseline specification (Figure 11a–c).

For `author_left_ratio`, the partial residuals display a clear convex pattern: residuals are systematically above zero at both ends of the distribution (near 0 and 1), and lower in the middle. This departure from linearity suggests that a linear term would be insufficient, and a spline basis is required to flexibly capture the curvature.

For `author_alignment_ratio`, we also observe a monotonic but curved relationship, with the slope increasing as the variable approaches 1. This pattern again motivates the use of a spline transformation rather than a single linear term.

In contrast, the partial residuals for `author_right_ratio` appear approximately linear across the [0,1] range, without strong evidence of systematic curvature. Therefore, we retain a simple linear effect for this predictor.

Given these patterns, we apply spline transformations (`bs` basis) to `author_left_ratio` and `author_alignment_ratio`, while modeling `author_right_ratio` linearly. We chose B-splines (`bs`) rather than natural cubic splines (`cr`), because the data are confined to the [0,1] interval and heavily used in interactions. In this setting, `bs` bases are more numerically stable due to their local support and reduced collinearity with intercept and linear terms, while `cr` tends to induce near-linear dependencies under boundary constraints.

Final Model Specification

Our final model specification includes:

- A main effect of `log_size` and its interaction with `platform`.
- `right_user_ratio` and spline-based transformation of `alignment_ratio` and `left_user_ratio`, interacted with both `log_size` and `platform`.
- Topic category and outlier status (influencer vs. non-influencer) as categorical moderators.

All models were estimated using the `statsmodels` package in Python, with robust standard errors and diagnostics to ensure stability.

Robustness to User Ideology Thresholds

In the main analysis, we defined user-level ideology using a threshold of 0.6 (i.e., a user is classified as left- or right-leaning if at least 60% of their posts are annotated as such; otherwise, they are labeled as center). To assess the robustness of our findings to this choice, we re-estimated Model 3 using alternative thresholds of 0.5 and 0.7. Table 6 reports the corresponding fit statistics and key parameters.

Across all three thresholds, the results remain substantively consistent. Pseudo-R^2 values are uniformly high, error metrics (MAE, RMSE) fluctuate only marginally, and, most importantly, the platform-size interaction term (Size$\times$TS) decreases markedly (even at the relatively noisy 0.5 threshold) in both the breadth and depth models once ideological structure is incorporated. This demonstrates that the explanatory power of user composition and alignment is robust to the choice of threshold used to classify user ideology. Accordingly, our central conclusion, that ideological structure accounts for cross-platform differences in cascade geometry, holds across reasonable operationalizations.

Robustness Checks: Alternative User Composition Specifications

To assess whether our findings depend on the way user composition is encoded, we compared four specifications: (1) left/right shares only, (2) majority/minority shares only, (3) left/right with alignment, and (4) majority/minority with alignment. Table 7 reports robust linear model (RLM) results for both cascade breadth and depth.

Including the alignment ratio substantially improves model fit for both outcomes (Breadth $R^2 \approx 0.95$; Depth $R^2 \approx 0.90$), confirming that interaction structure plays a central role in cascade geometry. The majority/minority framework slightly improves error metrics and consistently reduces the size of the $\log(\text{size}) \times$ platform interaction, suggesting that it absorbs part of the platform-specific compositional asymmetry. However, the overall improvement relative to the left/right specification is modest. Given that left/right distinctions are more theoretically grounded and more easily interpretable across platforms, we retain the left/right specification as our main model in the text, and report majority/minority results in the Appendix.

Marginal Effects of Ideological Structure

To further interpret Model 3, Figure 12 plots the marginal effects of alignment ratio, left-user ratio, and right-user ratio on cascade structure. For both depth and breadth, the results reveal that as cascade size increases, these relationships become less pronounced.

For **depth**, higher alignment is associated with shallower structures, consistent with the idea that ideologically homogeneous cascades are less likely to sustain extended exchanges. By contrast, cascades with fewer right-leaning participants exhibit greater layering, echoing our motif analysis finding that right-leaning users are less likely to construct fully right-aligned chains. Interestingly, the marginal effects of left- and right-user ratios are not symmetric: increases in right-leaning participation are linked to sharper declines in depth than comparable increases in left-leaning participation.

For **breadth**, the patterns diverge. Higher alignment corresponds to wider cascades, indicating that homogeneous groups are more likely to generate simultaneous uptake at the same level. The left-user ratio exhibits an inverted-U

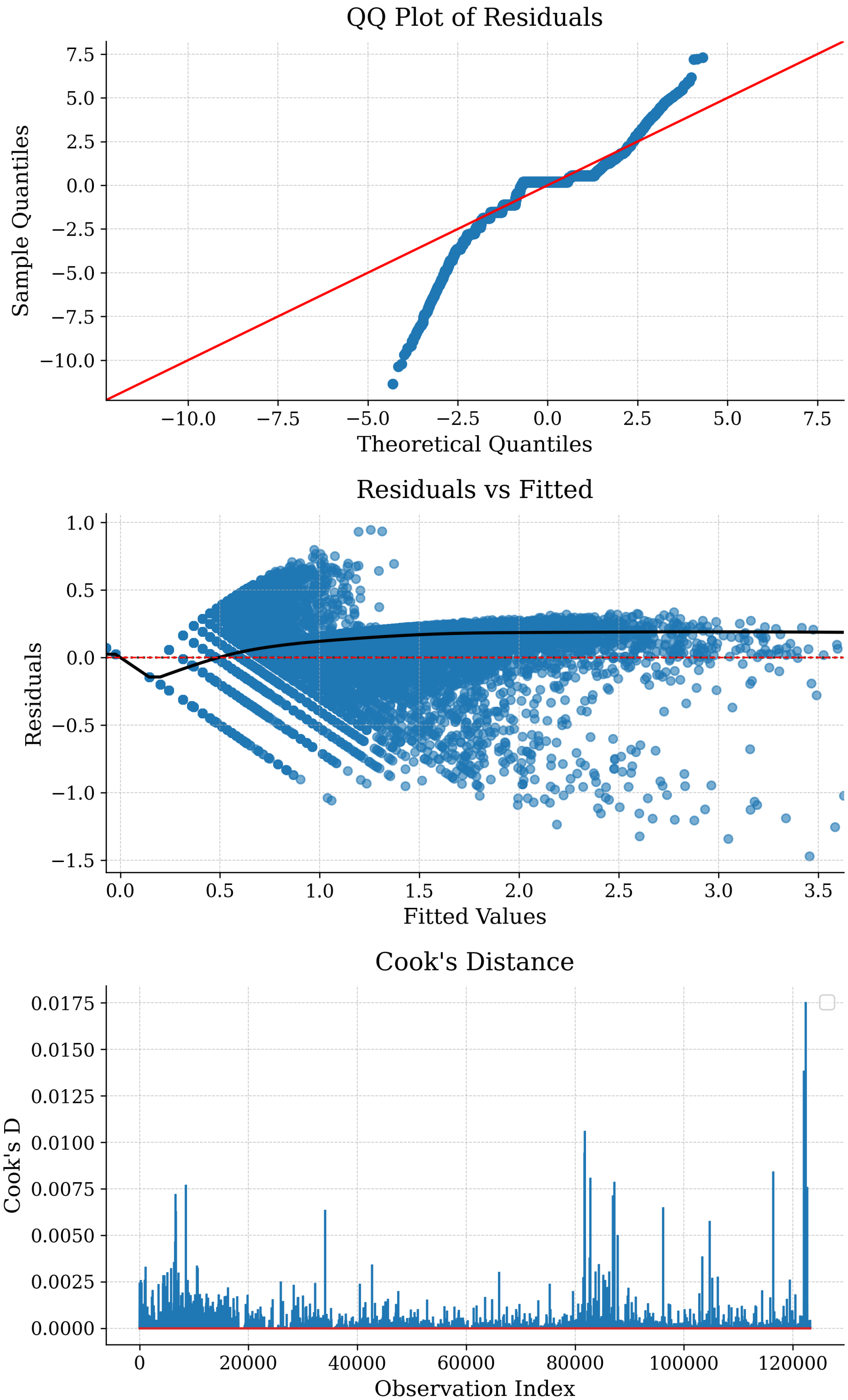

Figure 10: **Regression diagnostic plots assessing model assumptions.** The top panel shows a quantile–quantile (QQ) plot of the regression residuals, indicating substantial departure from normality, particularly in the tails. This suggests heavy-tailed error distribution, resembling a t-distribution. The bottom panel plots residuals against fitted values, revealing a fan-shaped pattern indicative of heteroskedasticity. A LOESS smoother (black line) is overlaid to highlight the non-constant variance pattern.

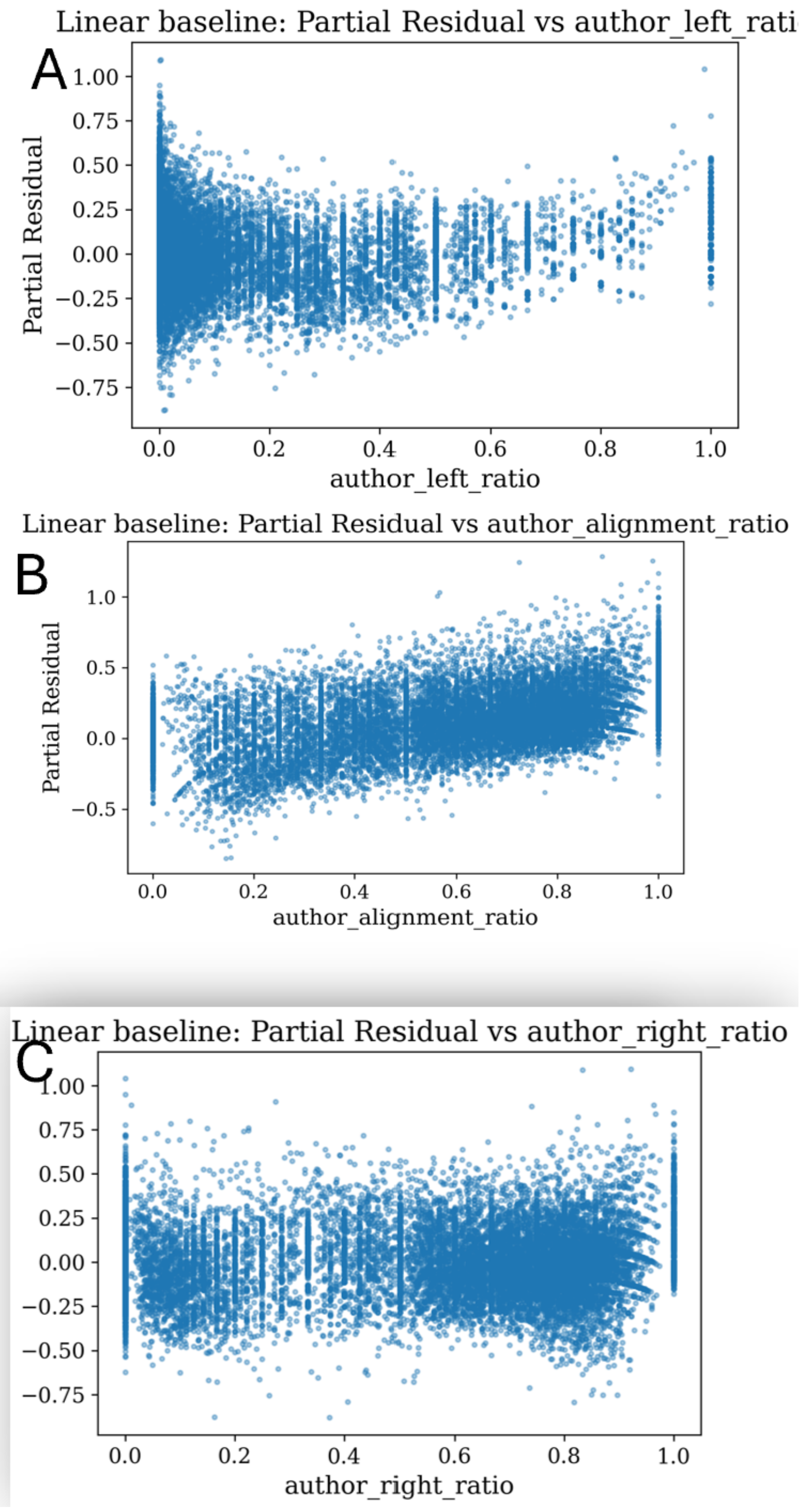

Figure 11: Partial residual plots under a linear baseline for (A) author_left_ratio, (B) author_right_ratio, and (C) author_alignment_ratio.

Table 6: Robustness check using alternative thresholds for defining user-level ideology. Results show that the main findings are stable across thresholds (0.5, 0.6, 0.7): platform gaps in cascade breadth and depth are consistently reduced once ideological structure is accounted for.

	Breadth (Log)				Depth (Log)			
Threshold	Pseudo-R^2	MAE	RMSE	Size$\times$TS	Pseudo-R^2	MAE	RMSE	Size$\times$TS
0.5	0.948	0.041	0.104	0.127	0.901	0.036	0.087	-0.143
0.6	0.952	0.042	0.101	0.023	0.905	0.036	0.085	-0.050
0.7	0.955	0.041	0.097	0.000	0.911	0.035	0.083	-0.000

Table 7: Comparison of user-composition encodings (Left/Right vs. Majority/Minority), with and without alignment. Robust Linear Models.

	Panel A: Composition only				**Panel B: + Alignment (df=3)**			
	Breadth (log_breadth)		**Depth (log_depth)**		**Breadth (log_breadth)**		**Depth (log_depth)**	
	L/R	Maj/Min	L/R	Maj/Min	L/R+Align	Maj/Min+Align	L/R+Align	Maj/Min+Align
Observations	123,189	123,189	123,189	123,189	123,189	123,189	123,189	123,189
Pseudo R^2	0.9312	**0.9322**	**0.8781**	0.8767	**0.9517**	0.9508	**0.9049**	0.9041
MAE	0.0565	**0.0563**	0.0445	**0.0431**	**0.0427**	0.0411	0.0368	**0.0359**
RMSE	0.1205	**0.1196**	**0.0964**	0.0970	**0.1010**	0.1019	**0.0852**	0.0855
$\beta_{\log(\text{size})}$	0.3646	0.3562	0.9002	0.8982	0.2526	0.4807	0.9286	0.8023
$\beta_{\log(\text{size})\times\text{platform(ts)}}$	0.2579	0.1073	-0.1950	-0.1112	0.0230	0.0719	-0.0547	-0.0552

Estimator note: entries use the default median-absolute-deviation scale and are reported without standard errors; the two encodings share the estimator, so the comparison between them is unaffected.

Notes: Left/Right (L/R) specification models the *left-user share* with a spline (df=3) and the *right-user share* linearly, interacted with $\log(\text{size})$ and platform. Majority/Minority (Maj/Min) re-codes composition by platform-dominant ideology (majority) vs. opposing (minority); the *majority share* is modeled with a spline (df=3) and the *minority share* linearly, with the same interactions. Panel B additionally includes the *alignment ratio* (spline, df=3) and its interactions with $\log(\text{size})$ and platform. Breadth—Maj/Min+Alignment overall error metrics (Pseudo R^2, MAE, RMSE) were not printed in the provided output and are marked as N/A; coefficient entries shown are from the reported summary. Full coefficient tables are available upon request.

relationship with breadth, with the widest cascades occurring at intermediate values. By contrast, the right-user ratio shows a more linear pattern: as right-leaning users become increasingly dominant, cascades grow consistently broader. This suggests that left- and right-leaning users across both platforms follow distinct modes of interaction.

Taken together, these marginal effects reinforce the regression evidence: it is not only platform size and topic content, but the **interaction between ideological composition and alignment** that systematically shapes cascade geometry. Homogeneity tends to flatten conversations into wide, shallow trees, while heterogeneity fosters greater layering in depth.

Network Motif Analysis Methodology

We use network motif analysis to characterize local interaction patterns in reply cascades. Network motifs are recurring subgraph configurations that appear more or less frequently than expected under a specified null model (Milo et al. 2002). In this study, motif analysis serves as a descriptive diagnostic of micro-level conversational structure rather than as an independent causal test of platform effects. The analysis evaluates whether specific 3-node ideological configurations are over- or under-represented relative to randomized reply trees that preserve the size of each observed cascade.

Motif Definition and Classification

We focus on directed 3-node tree motifs, the smallest nontrivial structures that distinguish branching from sequential reply patterns. Each motif is characterized by both its topology and the ideological labels of the participating users.

Topologically, we distinguish two motif families:

1. **Star motifs**: one parent node has two child nodes. In a reply tree, this corresponds to two replies attached to the same prior post.
2. **Chain motifs**: a directed path of length two, in which one reply receives a subsequent reply.

Each node is annotated with the ideology label of the corresponding user: Left (L), Center (C), or Right (R). Motifs are treated as ordered configurations. For chain motifs, order follows the direction of reply from the rootward node to the terminal node. For star motifs, the parent position is distinguished from the child positions, and ideological configurations are counted according to their ordered node positions in the enumerated tree motif. Thus, motif types capture both local reply structure and the positional arrangement of ideology within that structure.

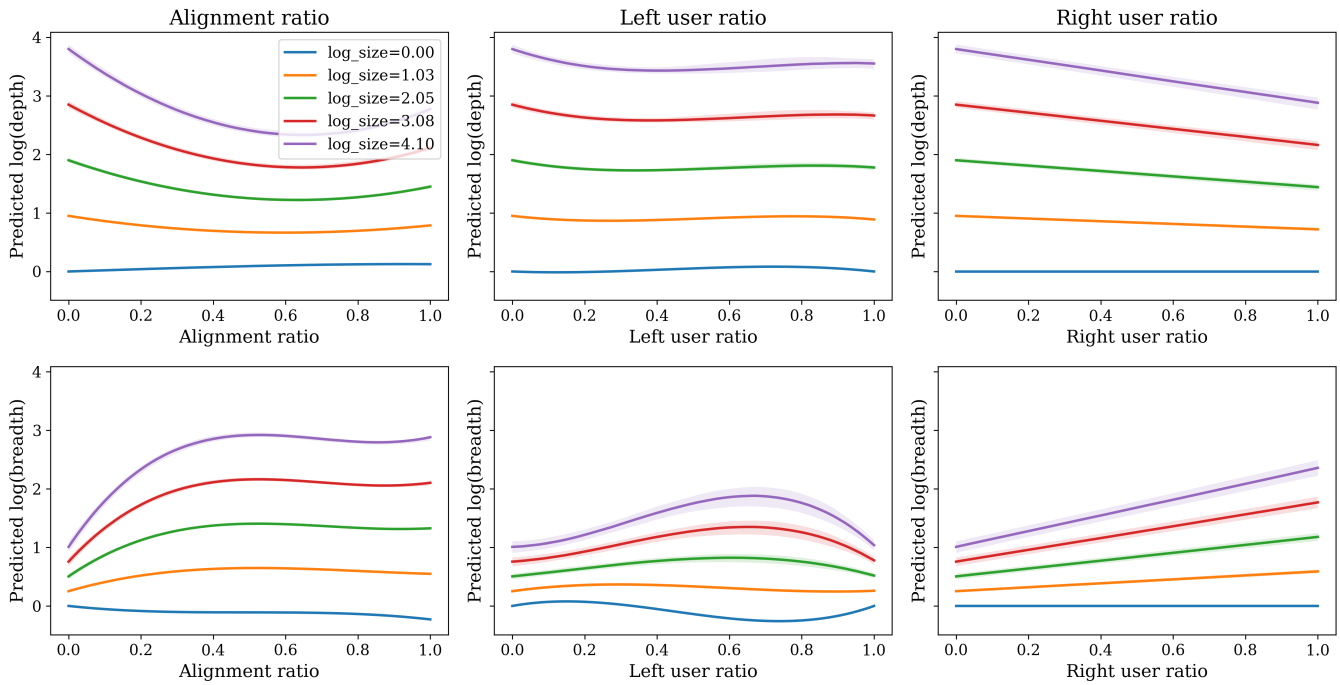

Figure 12: Marginal effects of ideological alignment, left-user ratio, and right-user ratio on cascade structure. Solid lines denote predicted log(depth) (top row) and log(breadth) (bottom row) at five equally spaced values of log(size) between its observed minimum and maximum. Shaded bands represent 95% bootstrap confidence intervals around the median prediction. Results indicate that homogeneous cascades (high alignment) tend to flatten, producing wide but shallow structures, whereas heterogeneous cascades deepen through sequential exchanges.

We also summarize motifs into broader interpretive families: *aligned* motifs, in which all three users share the same ideology; *one-different* motifs, in which two users share an ideology and the third differs; and *all-different* motifs, in which Left, Center, and Right users all appear in the same local structure. These families are used to interpret the full set of ordered motif results.

Counting Unit

Motifs are counted within each reply cascade tree and then aggregated by platform. Specifically, for each observed reply tree, we enumerate all connected 3-node tree subgraphs and classify each subgraph by topology and ordered ideological configuration. We then sum motif counts across all reply trees within the same platform. This procedure ensures that motifs represent local conversational configurations within observed threads, rather than triads formed by pooling users across unrelated conversations.

Null Model

To assess whether each motif appears more or less frequently than expected by chance, we compare the observed motif counts with a platform-specific ensemble of randomized reply trees. Randomization is performed separately within each platform and within each reply tree. For each observed tree, the null model preserves the number of nodes and edges. The only feature randomized is the assignment of edges among users within the tree. Node ideology labels are held fixed.

This null model therefore preserves cascade size and platform-specific ideological composition while randomizing the local attachment pattern among participants. It asks whether particular ideological configurations occur more or less often than expected given the observed set of users, their ideology labels, and the size of the reply trees. Because randomization is performed within trees, the comparison does not mix users across unrelated conversations or across platforms.

Randomization Procedure and Motif Scores

For each platform, we generate $R = 1{,}000{,}000$ randomized realizations of the reply trees. For each motif type i, we compute the observed motif count $N_{\mathrm{obs}}(i)$ and compare it with the distribution of counts obtained from the randomized trees. Let $\mu_{\mathrm{null}}(i)$ and $\sigma_{\mathrm{null}}(i)$ denote the mean and standard deviation of motif counts across the randomized realizations. We compute the standardized motif score as:

$$Z_i = \frac{N_{\mathrm{obs}}(i) - \mu_{\mathrm{null}}(i)}{\sigma_{\mathrm{null}}(i)}.$$

Positive values indicate that motif i appears more frequently than expected under the null model, whereas negative values indicate that it appears less frequently than expected.

Because the motif analysis is used as a descriptive diagnostic rather than as a confirmatory multiple-hypothesis testing procedure, we do not interpret individual motifs through binary significance thresholds. Instead, we report standardized deviations from the null distribution and focus on the relative direction and magnitude of motif over- or under-representation across platforms. This avoids treating the motif panel as a set of independent hypothesis tests and keeps the interpretation centered on recurring local interaction patterns.

Interpretation

The motif analysis identifies local ideological configurations that are amplified or suppressed relative to randomized reply trees with the same cascade sizes and participant ideology labels. It should not be interpreted as a causal estimate of why these structures arise. Instead, it complements the cascade-level models by showing which local interaction configurations contribute to broader platform differences in reply structure.

Platform Affordances and Technical Comparability

Table 8 sets out the affordances and technical lineage relevant to constructing comparable cascades. Cascade comparability does not imply exposure equivalence: ranking, moderation, default feeds, and visibility rules are partly opaque and cannot be identified from public traces, so this remains an observational comparison of interaction traces.

Computing Environment

All experiments were conducted on a single-node, multi-GPU system equipped with **8 NVIDIA H100 GPUs (80GB memory each)**. Model inference was deployed using **vLLM** with **bfloat16 (BF16) precision**, enabling efficient batched inference while maintaining numerical stability for long-context inputs.

Table 8: Platform affordances and technical lineage relevant to cascade construction.

Dimension	**Truth Social**	**Bluesky**
Technical lineage	Mastodon-derived platform, based on open-source social networking software	Built on the open-source AT Protocol ecosystem
Posting affordance	Supports user-authored posts	Supports user-authored posts
Reposting affordance	Supports repost-style resharing of posts	Supports repost-style resharing of posts
Reply affordance	Supports threaded replies	Supports threaded replies
Cascade observability	Public repost and reply traces are observable for collected posts	Public repost and reply traces are observable for collected posts
Feed and ranking	Platform-specific visibility and ranking may affect exposure; ranking process is not directly observed	Supports multiple feeds and feed-generator architecture; visibility may vary across selected or default feeds
Moderation and governance	Platform-specific moderation and visibility policies may affect observed interaction patterns	Platform-specific moderation, labels, and user/feed-level curation may affect observed interaction patterns
Analytic implication	Enables construction of repost and reply cascades, but the exposure process is not fully identified	Enables construction of repost and reply cascades, but the exposure process is not fully identified